

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/352643852>

A systematic review of resistance training methodologies for the development of lower body concentric mean power, peak power, and mean propulsive power in team-sport athletes

Article in *Sports Biomechanics* · June 2021

DOI: 10.1080/14763141.2021.1948601

CITATIONS

5

READS

500

3 authors:



Patrick Cormier

Canadian Sports Institute

7 PUBLICATIONS 63 CITATIONS

[SEE PROFILE](#)



Tomás T. Freitas

Universidad Católica de Murcia

107 PUBLICATIONS 1,133 CITATIONS

[SEE PROFILE](#)



Ken Seaman

University of New Brunswick

8 PUBLICATIONS 88 CITATIONS

[SEE PROFILE](#)

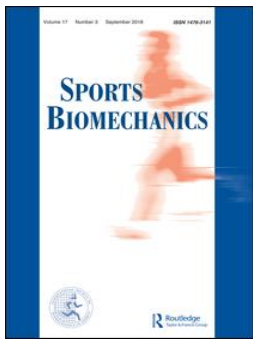
Some of the authors of this publication are also working on these related projects:



Sequence of Strength & Power exercises in team sports [View project](#)



External Load and Neuromuscular Performance in Elite Futsal Players: Seasonal Variations and the Relationship with Risk of Injury [View project](#)



A systematic review of resistance training methodologies for the development of lower body concentric mean power, peak power, and mean propulsive power in team-sport athletes

Patrick Cormier, Tomás T. Freitas & Kenneth Seaman

To cite this article: Patrick Cormier, Tomás T. Freitas & Kenneth Seaman (2021): A systematic review of resistance training methodologies for the development of lower body concentric mean power, peak power, and mean propulsive power in team-sport athletes, Sports Biomechanics, DOI: [10.1080/14763141.2021.1948601](https://doi.org/10.1080/14763141.2021.1948601)

To link to this article: <https://doi.org/10.1080/14763141.2021.1948601>



Published online: 04 Aug 2021.



Submit your article to this journal [↗](#)



View related articles [↗](#)



View Crossmark data [↗](#)



A systematic review of resistance training methodologies for the development of lower body concentric mean power, peak power, and mean propulsive power in team-sport athletes

Patrick Cormier ^a, Tomás T. Freitas ^{b,c,d} and Kenneth Seaman^a

^aFaculty of Kinesiology, University of New Brunswick, Fredericton, New Brunswick, Canada; ^bUCAM Research Center for High Performance Sport - Catholic University of Murcia, UCAM, Spain; ^cNAR - Nucleus of High Performance in Sport, São Paulo, Brazil; ^dFaculty of Sport Sciences, Catholic University of Murcia, UCAM, Murcia, Spain

ABSTRACT

This study aimed to systematically review training methods prescribed to develop lower-body power, determine their effectiveness for the development of lower-body mechanical power and their implementation in an annual training cycle amongst team-sport athletes. The absolute and relative outcome values of concentric mean power, peak power and mean propulsive power were extracted from 19 studies. Outcomes were assessed using baseline to post intervention percent change, effect sizes, and the level of evidence concerning the method's effectiveness. A thorough analysis of the literature indicated that, based on the high level of evidence, traditional (e.g., strength training alone) and combination training (e.g., complex and contrast) methods should be considered. Further, optimal load and velocity-based training can be implemented if coaches have access to the appropriate equipment to monitor movement velocity and mechanical power in every session. This is of particular importance in periods of the season where high volumes of technical-tactical training and congested fixture periods are present. Also, flywheel, eccentric overload and weightlifting methods have been shown to be effective although the level of evidence is low. Future research should expand on current training practices whilst adequately reporting actual training loads from sport-specific training and games alongside strength-power training protocols.

ARTICLE HISTORY

Received 2 February 2021
Accepted 22 June 2021

KEYWORDS

Evidence-based; force measurement; impulse; kinetics; performance

Introduction

The ability to generate 'power' is essential to the successful performance of neuromuscular and dynamic sport-specific tasks (Aagaard, 2003; Suchomel et al., 2016). Power, defined as the rate of performing mechanical work (Kawamori & Haff, 2004), is a result of neuromuscular actions that maximise the impulse produced (i.e., area under the force-time curve) which, in turn, determines the resulting velocity as a consequence of the impulse-momentum relationship (Winter et al., 2016). Based on previous work, the

underpinning aspect leading to performance in powerful muscular actions, such as jumping, is impulse (McBride et al., 2010). For instance, in a vertical jump, it is possible that jump height increases while power output decreases due to changes in jumping strategy; thus, vertical impulse (as explained by the impulse-momentum relationship) determines jump height. Nevertheless, power is still an important determinant and predictor of performance in sport-specific tasks (Cormie et al., 2011a), especially in team-sports, which often require players to perform a great deal of mechanical work in short timeframes.

To maximise performance, team-sport scientists and strength and conditioning coaches implement various training programs that target specific strength-power related qualities, hence optimising athletes' power production (McGuigan et al., 2012; Newton & Kraemer, 1994). Accordingly, greater maximal impulse capabilities are found with stronger individuals (Cormie et al., 2009; James et al., 2018). Along these lines, it is evident that diverse types of strength training interventions are effective in improving various neuromuscular qualities (e.g., rate of force development, stretch shortening cycle function, etc.). Moreover, and importantly, developing strength and power may lead to positive adaptations in actions that are specific to athletic performance (e.g., sprinting, jumping, changing of direction [COD], etc.) (Cormier et al., 2020; Freitas et al., 2017), while also enhancing the ability to tolerate weekly changes in training load (Malone et al., 2019), reducing injury rates (Lauersen et al., 2018; Malone et al., 2019), and increasing potentiation ability (Suchomel et al., 2016).

Although the aforementioned metrics are important, there are other power-related measures that can offer a more direct assessment of power corresponding to neuromuscular adaptations following athletic training (Dugan et al., 2004; Häkkinen, 1989). For example, concentric mean power (MP), peak power (PP), and mean propulsive power (MPP) have been extensively studied in the sport science literature (McMaster et al., 2014; Sanchez-Medina et al., 2010; Soriano et al., 2020). PP is characterised as the maximum instantaneous value achieved in the concentric phase of a movement (i.e., rate of doing work at an instant time) whereas MP is the average power produced during the whole concentric portion of an exercise (i.e., average rate of doing work over a time period considering both the 'propulsive' and 'braking' phases) (Sanchez-Medina et al., 2010; Sapega & Drillings, 1983). MPP is the concentric mean power output achieved in the concentric propulsive phase of a lift where the acceleration of the load is greater than the acceleration due to gravity (i.e., value without the decelerative 'braking' phase required to stop the movement) (Cronin & Sleivert, 2005; Sanchez-Medina et al., 2010). These are often measured with ballistic vertically oriented exercises such as unloaded or loaded jump squats at 0–60% 1 repetition-maximum (RM) or 20–80 kg (Loturco et al., 2017; Sanchez-Medina et al., 2010) but also with horizontal hip-dominant exercises like the hip thrust (Freitas et al., 2018; Ribeiro et al., 2020). Although all these values are associated with dynamic jumping tasks, relative PP and MPP may be more closely related to superior jump squat height in top-level athletes (Loturco et al., 2017). Further, it is important to consider that while relative values of vertical impulse are excellent predictors and determinants of jump height (nearly perfect correlations between both variables have been reported), relative MP, PP, and MPP values are also closely associated to jump performance, although these power values are dependent on jumping kinematics (i.e., jump strategy) (Kirby et al., 2011; McBride et al., 2010). This also applies to acceleration and deceleration in sprinting/

COD actions where horizontal-vertical braking and propulsive forces, and impulses must be applied within the constraints of short ground contact times (Dos'Santos et al., 2017; Harper et al., 2019; Kawamori et al., 2013; Loturco et al., 2020), as most athletic actions occur in timeframes < 300 ms (Aagaard, 2003; Turner et al., 2020). Thus, while keeping in mind the underpinning determinants of powerful muscular actions, concentric MP, PP, and MPP will be investigated for the purpose of this review.

Regarding training interventions, traditional resistance training alone (i.e., strength training with free weight, resistance machines, etc.) has long been shown to improve power (Häkkinen, 1989); however, many methods have emerged in the literature over the last few decades. For instance, plyometric and ballistic exercises (Chaabene et al., 2019), as well as weightlifting and its derivatives (Suchomel et al., 2015) have become popular as they approach the velocities achieved and forces applied in powerful athletic actions. Recent research has hypothesised that combining traditional resistance training with lighter load variations (i.e., plyometric, ballistic, and weightlifting derivatives) (Bauer et al., 2019; Cormier et al., 2020; Freitas et al., 2017; Marshall et al., 2021) may result in greater improvements in power over traditional methods alone. In addition, modern technologies have allowed a more precise prescription of exercises using autoregulation methods like velocity-based training (Guerriero et al., 2018; Mann et al., 2015; Weakley et al., 2021), where velocity in every repetition can be monitored (e.g., tracking barbell velocity in various exercises). Furthermore, training approaches using loads that maximise mechanical power can also be found and are often referred to as 'optimal load' or 'optimum power zone' training (Loturco, Nakamura et al., 2015; Soriano et al., 2015). Interestingly, methods emphasising the eccentric only (McNeill et al., 2019; Wagle et al., 2017) portion of a lift are also frequently implemented. Similarly, iso-inertial devices (e.g., flywheel) (Maroto-Izquierdo et al., 2019, 2017) have been used as an alternative approach to overload the eccentric phase of movements. Altogether, these methods used individually or in combination can lead to neural, structural, molecular, metabolic, and/or power performance adaptations in team-sport athletes (Cormie et al., 2011a).

Despite all the above-mentioned methods being used by practitioners for sport performance purposes, it may be challenging to know how or when to implement them during the season due to the constraints in 'real-world' team-sport settings. Therefore, investigating which methods can be the most effective to develop lower-body power is of interest to practitioners. As such, the purpose of this study was to (i) systematically review the literature on the methodologies used to develop MP, PP, and MPP in team-sports, (ii) identify which training methods are most effective at improving MP, PP, and MPP based on the level of evidence available in the literature, and (iii) consider the timing of the implementation of these training methods within a season or annual training plan.

Materials and methods

Experimental approach

Data sources and searches

The electronic search followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2009) and was conducted on the following databases: PubMed/Medline, SPORTDiscus, and Web of Knowledge (WoS). It

considered articles published in the English language before 15 October 2020. The following keywords were combined in the selected databases: ('resistance training' OR 'ballistic training' OR 'weightlifting' OR 'combination training' OR 'complex training' OR 'contrast training' OR 'velocity-based training' OR 'strength training' OR 'optimal load' OR 'plyometric') AND ('mean propulsive power' OR 'mean power' OR 'peak power' OR 'lower body' OR 'lower limb') AND ('performance' OR 'athletes' OR 'team-sport'). Reference lists from relevant articles were examined to find all possible papers that may fit the inclusion criteria.

Subjects

Eligibility criteria

Once the initial search was conducted, a reference manager software (Zotero, version 5.0.84, Corporation for Digital Scholarship, Vienna, VA, USA) was used to organise the imported studies into libraries. After the successful upload of all the articles searched in our databases, the duplicate items were identified and removed. Afterwards, the titles and abstracts were screened based on our eligibility criteria. The remaining full-text studies were assessed for inclusion.

Studies were considered for inclusion if: [1] at least one experimental group participated in an intervention with the goal of increasing lower-body neuromuscular power; [2] relative or absolute values of concentric MP, PP, or MPP were outcome measures; [3] participants mean age was >16 years and <45 years; [4] participants were from a competitive athletic team-sport population; [5] interventions lasted at least 6 weeks of continuous duration. Studies were excluded if: [1] they were not published in peer-reviewed journals or no full-text was available; [2] power values were derived from portable jump mats, optical measurement systems and/or timing gates due to their lack of emphasis on biomechanical aspects [3] concentric MP, PP, or MPP were not outcome measures; [4] participants had neuromuscular disorders or were taking any performance enhancing substances; and [5] interventions used unstable surfaces or were performed in hypoxic conditions. A threshold of >16 years of age was used due to the notion that strength-power training is often implemented to optimise movement competency and power performance adaptations from late puberty to adulthood (Meylan et al., 2014) in team-sports.

Procedures

Study selection and data extraction

Database searches were performed in an unblinded way, independently by one author (PC). After the removal of duplicates, abstract/titles were screened, and studies not related to the review's topic were excluded. The remaining articles were read and assessed for inclusion. Two reviewers selected the studies for inclusion (PC and KS), according to the eligibility criteria. If there were any that were unclear, the study was either omitted or included in the review when both authors achieved an agreement. One author (PC) extracted the following data from the selected studies: training background, country, sample size (n), sex, age, frequency, training duration, 1RM/BM ratio, seasonal period, training intervention, and the types of measurement that were used. In addition, all

baseline and post intervention data (including standard deviations) were extracted for all outcomes.

Quality assessment

The Physiotherapy Evidence Database (PEDro) scale was used to assess the methodological quality of each individual study included in this review (Maher et al., 2003). For each item, a score of '1' meaning the PEDro scale criteria was met, '0' score criteria not met, and '-' score not applicable. PEDro scores of 0–3 are considered 'poor', 4–5 'fair', 6–8 'good', and 9–10 'excellent' (Cashin & McAuley, 2020). Details on the PEDro scoring items are presented in Table 1.

Statistical analysis

This systematic review was carried out using Microsoft Excel 2016 (Microsoft Corporation, Redmond, WA, USA). In an excel form, relevant effects sizes (ES), 95% confidence intervals (CI), and pre-post intervention per cent change in means $[(\text{post mean} - \text{pre mean}) / \text{pre mean} * 100]$ were calculated if the data (pre-post means, standard deviation, and sample size) were available. If data were not available or unclear, the reported values from the individual studies were displayed. The ES (standardised mean difference) in each trial was calculated using

Table 1. Physiotherapy evidence database (PEDro) scores of the studies included in this review.

Study	Item 1*	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10	Item 11	Total
Cook et al. (2013)	1	1	1	1	-	-	-	1	1	1	1	7
Freitas et al. (2018)	1	1	1	1	-	-	-	1	1	1	1	7
Hermassi et al. (2019)	1	1	1	0	-	-	-	1	1	1	1	6
Horwath et al. (2019)	1	1	1	0	-	-	-	1	1	1	1	6
Helland et al. (2017)	1	1	1	1	-	-	-	1	1	1	1	7
Hammami et al. (2018)	1	1	1	1	-	-	-	1	1	1	1	7
Hansen et al. (2011)	1	1	1	1	-	-	-	1	1	1	1	7
Hermassi et al. (2011)	1	1	1	0	-	-	-	1	1	1	1	6
Ishida et al. (2020)	1	0	1	1	-	-	-	1	1	1	1	6
Loturco et al. (2016)	1	1	1	1	-	-	-	1	1	0	1	6
Los Arcos et al. (2014)	1	1	1	1	-	-	-	1	1	1	1	7
Loturco et al. (2013)	1	1	1	1	-	-	-	1	1	0	1	6
Núñez et al. (2018)	1	0	1	1	-	-	-	1	1	1	1	6
Otero-Esquina et al. (2017)	1	0	1	1	-	-	-	1	1	1	1	6
Rauch et al. (2018)	1	1	1	1	-	-	-	1	1	1	0	6
Ramirez et al. (2015)	1	0	1	1	-	-	-	1	1	0	1	5
Rønnestad et al. (2008)	1	1	1	1	-	-	-	1	1	1	1	7
Suarez-Arrones et al. (2018)	1	1	1	1	-	-	-	1	1	0	1	6
Talpey et al. (2016)	1	1	1	1	-	-	-	1	1	1	1	7

Item 1, eligibility criteria were specified; item 2, subjects were randomly allocated to groups; item 3, allocation was concealed; item 4, the groups were similar at baseline; item 5, there was blinding of all subjects; item 6, there was blinding of all therapists; item 7, there was blinding of all assessors; item 8, measures of at least one key outcome were obtained from more than 85% of the subjects initially allocated to groups; item 9, all subjects for whom outcome measures were available received the treatment or control condition as allocated or, data for at least one key outcome was analysed by 'intention to treat'; item 10, the results of between-group statistical comparisons are reported for at least one key outcome; item 11, the study provides both point measures and measures of variability for at least one key outcome. *Item 1 was not included as part of the study quality rating as it pertains to external validity. Total from a maximum of 10.

the Cohen's equations (Lakens, 2013). Hedges g correction was applied to Cohen's d to reduce small sample bias (Cohen, 1988). The following ES thresholds were used: <0.2 = trivial; $0.2-0.6$ = small; $>0.6-1.2$ = moderate; $>1.2-2.0$ = large; $>2.0-4.0$ = very large; and >4 = extremely large (Cohen, 1988; Hopkins et al., 2009).

Results

Search results

A total of 1474 articles were identified after the database search and removal of duplicates. Following title and abstract screening, 121 articles were left and assessed for inclusion based on our eligibility criteria. Further, after the exclusion of 102 studies with reasons, 19 were selected for assessment (Figure 1).

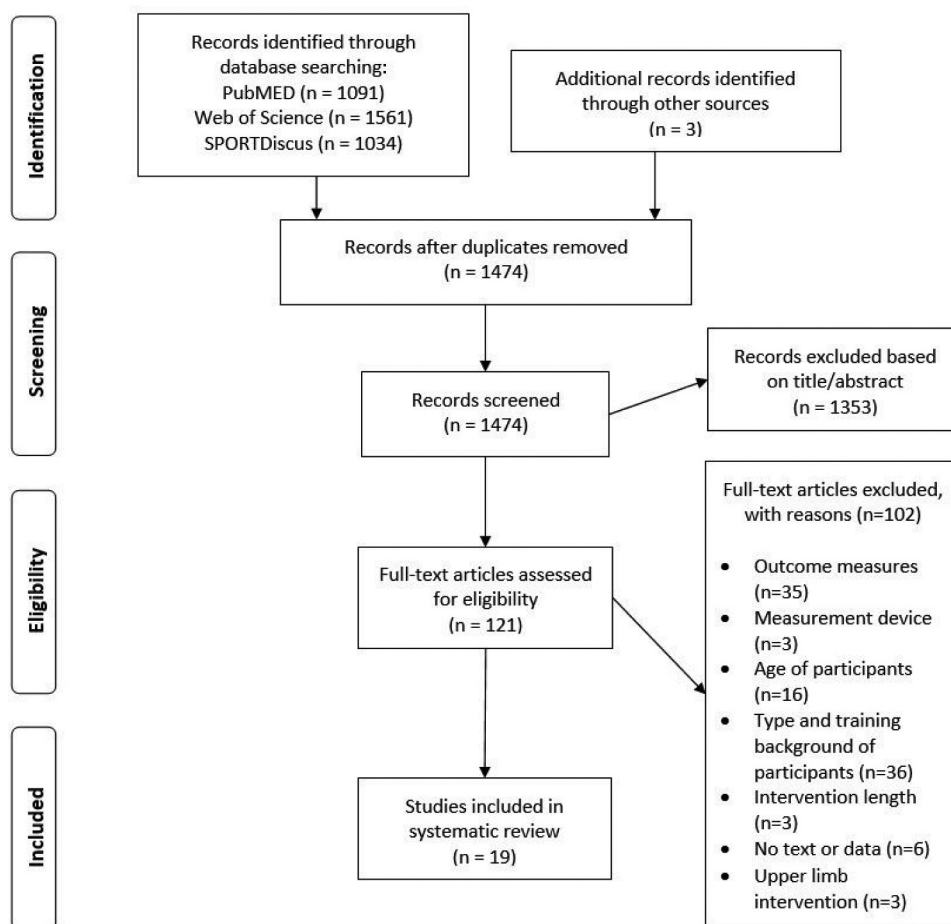


Figure 1. Flow diagram of the study selection process.

Methodological quality of studies

A mean quality score of 6 or above on the PEDro scale indicates the threshold for studies with low risk of bias. Altogether, the studies achieved the required standard for a low risk of bias (mean quality score = 6.37 ± 0.60) and obtained a score considered of ‘good’ quality (Table 1).

Study characteristics

Participants

A total of 19 studies met the inclusion criteria for the descriptive analysis portion of this review (Table 2). A total of 379 participants were in the experimental group, and 61 in control groups, making a total of 440 participants assessed in this study. The mean age of the participants ranged from 16 to 28 years and their strength levels (1RM/BM ratio) ranged from 1.2 to 2.4 in the back squat exercise. Individual studies originated from various countries with athletes from a variety of team-sports including soccer (Hammami et al., 2018; Ishida et al., 2020; Los Arcos et al., 2014; Loturco, Nakamura et al., 2016; Loturco et al., 2013; Otero-Esquina et al., 2017; Ramirez et al., 2015; Ronnestad et al., 2008; Suarez-Arrones et al., 2018; Talpey et al., 2016), handball (Hermassi et al., 2011, 2019), Australian rules football (Talpey et al., 2016), rugby (Cook et al., 2013; Hansen et al., 2011; Talpey et al., 2016), basketball (Freitas et al., 2018; Talpey et al., 2016), volleyball (Helland et al., 2017; Rauch et al., 2018), and ice hockey (Helland et al., 2017; Horwath et al., 2019). Noteworthy, Helland et al. (2017) randomly allocated badminton, ice hockey, and volleyball athletes into three distinct training methods. All participants had resistance training experiences of varying time spans. Specifically, some had several years (>1 year) of experience, and some studies only reported that participants had a sport-specific training background or stated that they were resistance trained (Hammami et al., 2018; Hansen et al., 2011; Hermassi et al., 2011, 2019; Ishida et al., 2020; Loturco, Nakamura et al., 2016; Loturco et al., 2013; Ramirez et al., 2015; Rauch et al., 2018; Ronnestad et al., 2008; Suarez-Arrones et al., 2018). Finally, participants were recruited from collegiate (Ishida et al., 2020; Rauch et al., 2018), amateur (Hammami et al., 2018; Helland et al., 2017; Nuñez et al., 2018; Talpey et al., 2016), semi-professional (Cook et al., 2013; Freitas et al., 2018; Horwath et al., 2019), and professional (Hansen et al., 2011; Hermassi et al., 2011, 2019; Los Arcos et al., 2014; Loturco, Nakamura et al., 2016; Loturco et al., 2013; Otero-Esquina et al., 2017; Ramirez et al., 2015; Ronnestad et al., 2008; Suarez-Arrones et al., 2018) team-sports.

Interventions

After all the studies were assessed for inclusion, the relevant studies were grouped into specific types of training (Table 3). Specifically, traditional (TDT) (Hammami et al., 2018; Hermassi et al., 2011; Ishida et al., 2020; Loturco, Nakamura et al., 2016; Ronnestad et al., 2008), combination (CBT) (Freitas et al., 2018; Hansen et al., 2011; Helland et al., 2017; Horwath et al., 2019; Los Arcos et al., 2014; Ronnestad et al., 2008; Talpey et al., 2016), velocity-based (VBT) (Ramirez et al., 2015; Rauch et al., 2018), optimal power load (OPL) (Freitas et al., 2018; Loturco, Nakamura et al., 2016; Loturco et al., 2013; Rauch et al., 2018), weightlifting and derivatives (WL)

Table 2. Participant characteristics from the studies included in this systematic review.

Study (year)	Level	Resistance training experience	Sport	Country	Season	Group	F (%)	Participants (n)	Age mean (±SD)	IRM/ BM ratio
Traditional strength training										
Hamami et al. (2018)	Amateur	Resistance trained	Soccer	Tunisia	In-season	EG CG	0	19 12	16.2 (0.6) 15.8 (0.2)	1.7
Hermassi et al. (2011)	Professional	Resistance trained (i.e., moderate training loads)	Handball	Tunisia	In-season	EG CG	0	12 12	22.1 (1.7) 20.2 (1.5)	1.9
Ishida et al. (2020)	Collegiate	-	Soccer	USA	Off-season	EG	0	17	19.6 (1.6)	-
Loturco et al. (2016)	Professional	Resistance trained (i.e., moderate-heavy training loads)	Soccer	Brazil	In-season	EG	0	12	23.1 (3.2)	-
Rønnestad et al. (2008)	Professional	Resistance trained	Soccer	Norway	Pre-season	EG CG	0	6 7	22 (2.5) 24 (1.5)	2.1
Combination training										
Freitas et al. (2018)	Semi-professional	>1 year resistance training	Basketball	Spain	In-season	EG	0	9	21.3 (4.3)	1.7
Horwath et al. (2019)	Semi-professional	>2 years resistance training	Ice Hockey	Sweden & Norway	-	EG	0	11	17.6 (0.9)	1.5
Helland et al. (2017)	Amateur	>2 years strength-power training	Badminton, volleyball, and ice hockey	Norway	Off-season	EG	26	13	20.0 (3.0)	1.4
Hansen et al. (2011)	Professional	Resistance trained	Rugby	Australia	Pre-season	EG EG	0	9 9	27.8 (4.5) 25.7 (4.5)	1.9
Los Arcos et al. (2014)	Professional	>3 years strength training	Soccer	Spain	Pre-season (5-weeks), In-season (3-weeks)	EG EG	0	11 11	20.3 (1.9) 19.6 (1.6)	-

(Continued)

Table 2. (Continued).

Study (year)	Level	Resistance training experience	Sport	Country	Season	Group	F (%)	Participants (n)	Age mean (±SD)	1RM/ BM ratio
Rønnestad et al. (2008)	Professional	Resistance trained	Soccer	Norway	Pre-season	EG CG	0	8 7	23 (2) 24 (1.5)	2.4
Talpey et al. (2016)	Amateur	>1 year resistance training	Australian rules football, rugby union, basketball, and soccer	Australia	-	EG	0	9	21.4 (3.5)	1.7
Velocity-based training										
Rauch et al. (2018)	Collegiate	-	Volleyball	USA	Off-season	EG	100	15	19.3 (1.4)	-
Ramirez et al. (2015)	Professional	-	Soccer	Spain	-	EG	0	18	23.4 (3.7)	1.6
Optimal power load										
Freitas et al. (2018)	Semi-professional	>1 year resistance training	Basketball	Spain	In-season	EG	0	9	21.3 (4.3)	1.7
Loturco et al. (2016)	Professional	Resistance trained (i.e., moderate-heavy training loads)	Soccer	Brazil	In-season	EG	0	11	23.9 (4.4)	-
Loturco et al. (2013)	Professional	Resistance trained	Soccer	Brazil	Pre-season	EG	0	16 16	19.2 (0.7) 19.1 (0.7)	1.2
Rauch et al. (2018)	Collegiate	-	Volleyball	USA	Off-season	EG	100	15	19.3 (1.4)	-
Weightlifting and derivatives										
Hermassi et al. (2019)	Professional	-	Handball	Tunisia	In-season	EG CG	0	11 11	20.7 (0.5)	1.9
Helland et al. (2017)	Amateur	>2 years strength-power training	Badminton, volleyball, and ice hockey	Norway	Off-season	EG	26	13	20.0 (3.0)	1.4
Eccentric overload training										
Horwath et al. (2019)	Semi-professional	>2 years resistance training	Ice hockey	Sweden & Norway	-	EG	0	11	18.4 (0.7)	1.5

(Continued)

Table 2. (Continued).

Study (year)	Level	Resistance training experience	Sport	Country	Season	Group	F (%)	Participants (n)	Age mean (±SD)	1RM/BM ratio
Flywheel training										
Núñez et al. (2018)	Amateur	-	Various team sport players	Spain	-	EG (UG) EG (BG)	0	14 13	22.8 (2.9) 22.6 (2.7)	-
Mixed training										
Cook et al. (2013)	Semi-professional	>2 years resistance training	Rugby	UK	Pre-season	EG	0	5	19.4 (0.5)	1.6
Helland et al. (2017)	Amateur	>2 years strength-power training	Badminton, volleyball, and ice hockey	Norway	Off-season	EG	26	13	19.8 (0.8) 20.0 (3.0)	1.4
Otero-Esquina et al. (2017)	Professional	>5 years of sessions (4–5) of combined soccer & strength-power & 1 match/week over 10 months/year	Soccer	Spain	In-season	EG EG CG	0	12 12 12	17 (1)	-
Suarez-Arnes et al. (2018)	Professional	-	Soccer	Italy	In-season	EG	0	14	17.5 (0.8)	-

Data is represented in mean (±SD), n, or range. EG = experimental group, CG = control group, UG = unilateral group, BG = Bilateral group, 1RM = 1 repetition-maximum, BM = body mass, F = female
, ' = data not presented in study

Table 3. Intervention characteristics and results from studies included in this systematic review.

Study (year)	Intervention	Intensity	Duration (wks)	Freq/week	Instrumentation	Type of measurement (units)	Results (ES = g (95% CI), % = pre-post change)
Traditional strength training							
Hannami et al. (2018)	HS	70–90%1RM	8	2	Force platform	PP in CMJ (Watts), Relative PP in CMJ (W/kg), PP in SJ (W), Relative PP in SJ (W/kg)	ES = 1.13 (0.45–1.82), % = 26.7; ES = 1.36 (0.65–2.07), % = 23.1; ES = 0.67 (0.01–1.32), % = 12.1; ES = 0.87 (0.20–1.53), % = 9.0 ^a
Hermassi et al. (2011)	HS	80–95%1RM	8	2	Force platform	PP in CMJ (Watts), Relative PP in CMJ (W/kg), PP in SJ (W), Relative PP in SJ (W/kg)	ES = 0.93 (0.08–1.77), % = 15.3; ES = 2.33 (1.29–3.37), % = 13.6; ES = 0.14 (–0.67–0.94), % = 3.3; ES = 0.20 (–0.60–1.00), % = 3.4 ^a
Ishida et al. (2020)	BS, DB lunge, clean pull, deadlift	65–95%1RM	7	3	Force platform	PP in SJ, 20, 40 kg (W)	ES = 0.17 (–0.51–0.84), % = 3.6; ES = 1.75 (0.96–0.99), % = 47.0; ES = 0.31 (–0.37–0.99), % = 6.8 ^a
Loturco et al. (2016)	HS and JSs	HS (60–90%1RM), JS (30% 1RM)	6	3	Linear position transducer	MPP in JS, BM & 40%BM (W)	ES = 1.50, % = 8.3; ES = 0.25, % = 3.0 ^b
Romestad et al. (2008)	HS and hip flexion exercises	85–90%1RM	7	2	Force platform	PP in SJ 20, 35, 50 kg (W)	ES = 3.07 (1.40–4.74), % = 9.9; ES = 2.38 (0.90–3.86), % = 7.4; ES = 3.83 (1.93–5.74), % = 11.1 ^a
Combination training							
Freitas et al. (2018)	HS paired with HS (OL), and hip thrust paired with hip thrust (OL)	80%1RM and OL	6	2	Force platform	PP in CMJ (W)	ES = 0.22 (–0.70–1.15), % = 3.9 ^a
Horwath et al. (2019)	Bilateral and unilateral JSs	20–60%PP (unilateral), 10–20%PP (bilateral)	8	2–3	Linear position transducer	MP in JS, mean across loads 20, 40, 60, 80 kg (W)	ES = 0.97 (0.09–1.85), % = 17.5 ^a
Helland et al. (2017)	Bilateral and unilateral squats, and CMJs	Squats (85–95%1RM), CMJ (10–60%1RM)	8	2	Linear position transducer	PP in CMJ (W)	ES = small (0.20–0.59), % = 8.1 ^b

(Continued)

Table 3. (Continued).

Study (year)	Intervention	Intensity	Duration (wks)	Freq/week	Instrumentation	Type of measurement (units)	Results (ES = g (95% CI), % = pre-post change)
Hansen et al. (2011)	F5s, clean pulls, box squats, power cleans, BS, and JSs. Note: traditional or in clusters	80–95%1RM, JS (10–20% 1RM)	8	2	Force platform	PP in JS 0, 20, 40, 60 kg (W)	ES = 0.30 (–0.63–1.23), % = 3.8; ES = 0.27 (–0.66–1.20), % = 2.9; ES = 0.82 (–0.14–1.78), % = 7.2; ES = 0.52 (–0.42–1.46), % = 4.4; ES = 0.18 (–0.74–1.11), % = 2.0; ES = 0.38 (–0.55–1.31), % = 4.8; ES = 0.04 (–0.88–0.96), % = 0.5; ES = 0.17 (–0.76–1.10), % = 2.7 ^a ES = 0.79 (–0.08–1.65), % = 13.5; ES = 0.61 (–0.25–1.46), % = 12.2 ^a
Los Arcos et al. (2014)	Vertical: Unilateral and bilateral HSs, gastrocnemius, CMJ, CMJ to box, Unilateral CMJ to box, VJ with load (5%BM), skipping VJ, unilateral DJ; Vertical/horizontal: Unilateral and bilateral HSs, gastrocnemius, sled walking, hip external wall drill, double triple jump, hip external wall drill, unilateral DJ, skipping HJ	30–35%PP (unilateral), 70–74%PP (bilateral)	8	2	Linear position transducer	PP in HS (kg at 1 m·s ^{–1})	ES = 2.21 (0.97–3.46), % = 9.9; ES = 1.71 (0.57–2.86), % = 8.0; ES = 1.68 (0.54–2.82), % = 9.5 ^a ES = 0.16 (–0.77–1.08), % = 4.0; ES = –0.02 (–0.94–0.90), % = –0.5; ES = 0.07 (–0.86–0.99), % = 1.6; ES = 0.19 (–0.65–1.02), % = 3.9; ES = –0.18 (–1.02–0.66), % = –3.5; ES = –0.03 (–0.87–0.80), % = –0.6 ^a
Rønnestad et al. (2008)	HS, hip flexion exercise, leg bounds, bilateral hurdle jumps, and unilateral forward hops	85–90%1RM	7	2	Force platform	PP in SJ 20, 35, 50 kg (W)	ES = 0.16 (–0.77–1.08), % = 4.0; ES = –0.02 (–0.94–0.90), % = –0.5; ES = 0.07 (–0.86–0.99), % = 1.6; ES = 0.19 (–0.65–1.02), % = 3.9; ES = –0.18 (–1.02–0.66), % = –3.5; ES = –0.03 (–0.87–0.80), % = –0.6 ^a
Talpey et al. (2016)	HSs and JSs	80%1RM, BM	9	2	Force platform & linear position transducer	PP in CMJ, unloaded (W), PP in CMJ, 10% 1RM (W), PP in CMJ, 20%1RM	ES = 0.86, % = 18.3 ES = 1.32, % = 15.7 ^b
Velocity-based training							
Rauch et al. (2018)	BS and dead lifts	0.55–0.70 m·s ^{–1} , 0.85–1.0 m·s ^{–1}	7	3	Linear position transducer	PP in BS and deadlift (W)	ES = 0.86, % = 18.3 ES = 1.32, % = 15.7 ^b
Ramirez et al. (2015)	HS and hip thrusts	65–75%1RM whilst maintaining within ± 10% movement velocity	10	2	Linear position transducer	Relative PP (W/kg), PP in HS (W)	ES = 0.81 (0.13–1.49), % = 12.6; ES = 0.89 (0.21–1.58), % = 12.6 ^a
Optimal power load							
Freitas et al. (2018)	HS and hip thrusts	Individual OL	6	2	Force platform	PP in CMJ (W)	ES = 0.15 (–0.77–1.08), % = 2.9 ^a

(Continued)

Table 3. (Continued).

Study (year)	Intervention	Intensity	Duration (wks)	Freq/week	Instrumentation	Type of measurement (units)	Results (ES = g (95% CI), % = pre-post change)
Loturco et al. (2016)	JSs	Individual OL	6	3	Linear position transducer	MPP in JS, BM & 40%BM (W)	ES = 1.70, % = 14.5; ES = 1.20, % = 13.0 ^b
Loturco et al. (2013)	BS and JSs	30–80%1RM, $\pm 15\%$ (ranging from 30 to 60% 1RM) JS OL. Note: using a velocity vs intensity-based approach	6	2	Linear position transducer	MP in BS, 60%1RM (W), MPP in JS, 45%1RM (W)	Velocity vs Intensity: BS % = 18.5 vs 20.4; MPP JS % = 29.1 vs 31.0 ^b
Rauch et al. (2018)	BS and dead lifts	0.85–0.9 m·s ⁻¹	7	3	Linear position transducer	PP in BS and deadlift (W)	ES = 0.79, % = 19.8 ES = 1.77, % = 20.1 ^b
Weightlifting and derivatives							
Hermassi et al. (2019)	Snatches, clean and jerk	55–75%1RM	12	2	Force platform	Relative PP in CMJ (W/kg), Relative PP in SJ (W/kg)	ES = 0.58 (–0.28–1.43), % = 9.2; ES = 1.18 (0.28–2.09), % = 20.3 ^a
Helland et al. (2017)	Clean with front squat, hang clean, snatch, power jerk	85–95%1RM	8	2	Linear position transducer	PP in CMJ (W)	ES = trivial (<0.20), % = 2.6 ^b
Eccentric overload training							
Horwath et al. (2019)	Iso/ecc: Unilateral and bilateral Squats & JS	Squats (0.4 ms ⁻¹), JS (10–60% 1RM + 120–140% ecc.)	8	2–3	Linear position transducer	MP in JS, mean across loads 20, 40, 60, 80 kg (W)	ES = 0.76 (–0.10–1.63), % = 12.5 ^a
Flywheel training							
Núñez et al. (2018)	HSs using Kbox	0.05 (unilateral), 0.10 (bilateral) kg/m ² moment inertia	6	2	Rotary position transducer	Relative PP (HSs 0.05 kg/m ² , 0.10 kg/m ² moment inertia)	ES = 0.68 (–0.08–1.44), % = 19.4; ES = 1.38 (0.53–2.24), % = 36.9 ^a
Mixed training							
Cook et al. (2013)	Squats, mid-thigh pull, Romanian deadlifts with overspeed assisted CMJs and downhill sprints (40 m)	Eccentric group: 120%1RM, no conc phase; Traditional group: 80%1RM; Overspeed exercises: CMJs with reduced BM by 20%	12	2	Force platform	PP in CMJ, unloaded (W)	ES = 0.94 ^b

(Continued)

Table 3. (Continued).

Study (year)	Intervention	Intensity	Duration (wks)	Freq/week	Instrumentation	Type of measurement (units)	Results (ES = g (95% CI), % = pre-post change)
Helland et al. (2017)	Bilateral and unilateral squats and CMJs	Squats (0.2–0.4 m·s ⁻¹), CMJ (10–20%1RM+120–140% ecc)	8	2	Linear position transducer	PP in CMJ (W)	ES = small (0.20–0.59), % = 6.1 ^b
Otero-Esquina et al. (2017)	Full squat, YoYo leg curls, plyometrics (tuck jumps, DJs, hurdle jumps), and resisted sprints	40–55%1RM, FS (MPV = ~1.28–1.07 m·s ⁻¹)	7	1–2	Force platform	Relative PP in CMJ (W/kg)	ES = 0.80 (–0.03–1.63), % = 7.2; ES = 0.40 (–0.40–1.21), % = 4.9 ^a
Suarez-Arrones et al. (2018)	Single leg Yo-Yo leg curl, single leg hip extensions, lunges, HS variations using Kbox, lateral HS in Kbox, Russian belt hip rotation, and ankle extensions	0.05 & 0.10 kg/m ² moment inertia (Kbox)	27	2	Linear position transducer	MPP in HS, 30 and 40 kg (W)	ES = 0.45–1.73; % = LL 30 kg = 4.9, LL 40 kg = 13.7, RL 30 kg = 3.0, RL 40 kg = 11.5, Bilateral 30 kg = 3.3, Bilateral 40 kg = 11.5 ^b

Data is represented in mean (±SD), n, or range. CMJ = countermovement jump, FS = front squat, BS = back squats, HS = half squat, SJ = squat jump, JS = jump squats, VJ = vertical jump, DJ = drop jump, DB = dumbbell, BM = body mass, RM = repetition maximum, ES = effect size, CI = confidence interval, g = hedges g, OL = optimal load, Iso = isometric, ecc = eccentric, MP = mean power, MPP = mean propulsive power, PP = peak power, LL = left leg, RL = right leg, MPV = mean propulsive velocity, W = Watts, kg = kilograms

^aPre-post data was extracted, ES, 95% CI, and % change calculated by the authors of this review.

^bResults data extracted from the study due to no pre-post data presented.

(Helland et al., 2017; Hermassi et al., 2019), eccentric overload (EOT) (Horwath et al., 2019), flywheel (FW) (Nuñez et al., 2018), and mixed (MT) (Cook et al., 2013; Helland et al., 2017; Otero-Esquina et al., 2017; Suarez-Arrones et al., 2018) training. If a study compared two distinctly different methods of training (i.e., two or more intervention groups), the experimental groups were divided into their respective training category for the purpose of this review. Notably, there were no studies that met the inclusion criteria regarding plyometric training. TDT training interventions lasted from 6 to 8 weeks, from 2 to 3 times/week, using loads ranging from 30% to 95%1RM. CBT studies were 6 to 9 weeks duration, from 2 to 3 times/week frequency. Regarding the loads for CBT, a variety was used from 10% to 95%1RM combining higher-load strength and lower-load power exercises. One CBT study (Freitas et al., 2018) used a modified complex training method using moderate load (80%1RM) strength exercises paired with OPL conditions in combination. Within the VBT interventions, interventions from 7 to 10 weeks, and 2 to 3 times/week were used. Velocity ($\text{m}\cdot\text{s}^{-1}$) was used to monitor the load in different formats: velocity thresholds (e.g., $0.85\text{--}1.0\text{ m}\cdot\text{s}^{-1}$) and placing constraints (e.g., not allowing velocity $\pm 10\%$ with a corresponding load). The OPL training interventions lasted 6 to 7 weeks, at a frequency of 2 to 3 times/week at intensities of $0.85\text{--}0.9\text{ m}\cdot\text{s}^{-1}$, $1\text{ m}\cdot\text{s}^{-1}$, or the load that maximised mechanical power and loads ranging from 30% to 80%1RM. Concerning WL, the interventions lasted 8 to 12 weeks with a frequency of 2 times/week (loads ranged from 55% to 95%1RM). The EOT intervention was 8 weeks in duration, 2 to 3 times/week with overloads of 120–140%1RM. The FW training intervention lasted 6 weeks and was performed 2 times/week, using 0.05 (unilaterally) and 0.10 (bilaterally) kg/m^2 moment inertia using the Exxentric kbox (Exxentric AB, Stockholm, Sweden). MT durations ranged from 7 to 27 weeks in duration, 1–2 times/week, using many different loading protocols (mixed protocols where >2 training types were used). From the included studies, 3 were performed in the off-season (Helland et al., 2017; Ishida et al., 2020; Rauch et al., 2018), 4 in the pre-season (Cook et al., 2013; Hansen et al., 2011; Loturco et al., 2013; Ronnestad et al., 2008), 7 in the in-season (Freitas et al., 2018; Hammami et al., 2018; Hermassi et al., 2011, 2019; Loturco, Nakamura et al., 2016; Otero-Esquina et al., 2017; Suarez-Arrones et al., 2018), and 1 study in the pre and in-season (Los Arcos et al., 2014) periods.

Outcome measures

The instruments used to measure power outcomes in the studies included were force platforms (Cook et al., 2013; Freitas et al., 2018; Hammami et al., 2018; Hansen et al., 2011; Hermassi et al., 2011, 2019; Ishida et al., 2020; Otero-Esquina et al., 2017; Ronnestad et al., 2008; Talpey et al., 2016) and linear position transducers (Helland et al., 2017; Horwath et al., 2019; Los Arcos et al., 2014; Loturco et al., 2013; Nuñez et al., 2018; Ramirez et al., 2015; Rauch et al., 2018; Suarez-Arrones et al., 2018; Talpey et al., 2016) (Table 3). MP, PP, and MPP values derived from data collected by force platforms are considered a kinetic measurement method and linear position transducers are considered a kinematic method (Cormie et al., 2007). Force platforms obtain data via ground reaction forces and velocity of centre of mass as integrated from the impulse-

momentum theorem and acceleration-time curve (Dugan et al., 2004; McMaster et al., 2014; Soriano et al., 2020). Linear position transducers obtain data from the mass of the system (with or without the inclusion of BM of the athlete in the calculation) and velocity of the position of the transducer attachment point on the barbell as differentiated by the displacement-time data (McMaster et al., 2014; Soriano et al., 2020) which is subsequently analysed with software to derive power values (Harris et al., 2010). A rotary position transducer was used in one study (Nuñez et al., 2018) which employs the same concept as linear position transducers to obtain power values. Furthermore, power values were obtained in one study (Talpey et al., 2016) using synchronised force platforms and linear position transducer (combined kinetic-kinematic method) via the instantaneous product of vertical ground reaction force and bar velocity. It is important to note that obtaining power measures combining ground reaction force (from force platforms) and barbell velocity may overestimate velocity and power since the barbell may undergo displacement at a greater velocity compared to the centre of mass of the athlete (Lake et al., 2012). Further, sport scientists and strength and conditioning specialists may employ different strategies to calculate power values depending on the type of sport, loading parameters (barbell load) and selected exercise (e.g., squat vs jump squat) whereby some may be interested in using barbell power (including barbell mass only), body power (BM without barbell), or system power (including BM + barbell) (McBride et al., 2011). Practically speaking, using the barbell power method could be relevant for athletes who must primarily maximise power production against external resistances in their sport (e.g., throwers or competitive weightlifters). Conversely, using body power or system power may be more suitable for team-sport athletes (Balsalobre-Fernández & Torres-Ronda, 2021; McBride et al., 2011). Further, when assessing power values in a back squat vs a jump squat, the displacement of the centre of mass of the athlete in the propulsive phase of the movement would be greater in the latter than in the former, thus resulting in a higher power output in the jumping task (McBride et al., 2011). Whilst keeping this in mind, power was measured in the half-squat (HS) (Los Arcos et al., 2014; Nuñez et al., 2018; Ramirez et al., 2015; Suarez-Arrones et al., 2018), back squat (BS) (Loturco et al., 2013; Rauch et al., 2018), deadlift (Rauch et al., 2018), jump squat (JS) (Hansen et al., 2011; Horwath et al., 2019; Loturco, Nakamura et al., 2016; Loturco et al., 2013), countermovement jump (CMJ) (Cook et al., 2013; Freitas et al., 2018; Hammami et al., 2018; Helland et al., 2017; Hermassi et al., 2011, 2019; Otero-Esquina et al., 2017; Talpey et al., 2016), and squat jump (SJ) (Hermassi et al., 2011, 2019; Ishida et al., 2020; Ronnestad et al., 2008). Tests were performed unloaded or with a range of loads and the outcome values were recorded as concentric MP, PP, or MPP (Table 3).

Based on the studies and outcomes analysed, Table 4 summarises the potential of the different methods for increasing lower-body power measures according to the level of evidence available in the literature to date. The level of evidence was determined by a combined assessment of the ES reported in each individual study, the amount of evidence present (i.e., number of studies and total sample size), and the expert judgement of the authors of this review.

Table 4. Level of evidence relating to the potential effectiveness of a training method for improving power.

Training method	Level of evidence for the effectiveness of a training method potentially improving lower-body power (i.e., MP, PP, and MPP) ^a
Traditional strength	+++++
Combination	+++++
Velocity-based	+++
Optimal power load	++++
Weightlifting and derivatives	++
Eccentric overload	+
Flywheel	+
Mixed	++++
Plyometric	*

MP = mean power, PP = peak power, MPP = mean propulsive power

^aTraining methodology classified on scale from +, meaning low evidence, and ++++++, meaning high evidence; *means lack of evidence.

Discussion and implications

The aims of this study were to systematically review the scientific literature on the methodologies used to develop concentric MP, PP, and MPP in team-sports, identify which training methods may be the most effective based on the level of evidence available, and consider the timing of the implementation of these training methods within a season or annual training plan. Importantly, for a better understanding of the discussion section, it must be recognised that the measurement instrument and type of exercise used may affect outcome values. For example, when assessing PP in the JS using a linear position transducer, the values reported usually indicate the power developed ‘on the barbell’ (i.e., only the external load is considered for power output determination) (Loturco, Pereira et al., 2016; Loturco, Suchomel, Bishop et al., 2018; Loturco, Suchomel, James et al., 2018). Conversely, calculating PP in the JS (with or without added external loads) using force platforms may yield completely different power outputs since, in this case, the system mass (BM + external load) is utilised in the computation. It is also important to keep in mind that the differences in movement patterns (e.g., squat depth), loading parameters (e.g., no load vs incremental loading), methodology (e.g., data collection and processing technique), and athlete’s competitive level (e.g., amateur vs elite) or training status (e.g., strong vs weak) may affect power outcome measures (McMaster et al., 2014). This being said, the present results indicate that all training methodologies included in this review effectively improve the expression of lower-body mechanical power production amongst team-sport athletes whilst performing sport-specific training concurrently. Further, the information from this systematic review may have important implications for practitioners when making decisions on which methods should be used within the context of a team-sports season.

Traditional strength training

TDT methods, in our analysis, resulted in trivial-very large improvements. This method has been extensively researched and has been found to be effective in increasing muscular power (Aagaard, 2003). Further, positive associations between maximal dynamic strength and sprint performance (Wisløff et al., 2004) and reductions in injury risk

(Malone et al., 2019) have been reported in team-sports, supporting the implementation of TDT programmes. Therefore, the level of evidence concerning its use is high. Accordingly, looking specifically at a study by Ronnestad et al. (2008) with professional soccer players, greater effects were achieved using traditional methods rather than using CBT, although they both demonstrated large effects. Interestingly, only the CBT group reported significant improvements in sprint speed, thus it appears that TDT may lead to PP improvements, while CBT may have a greater transfer to velocity-specific performance-based adaptations, possibly due to inclusion of more rapid movement velocities with the latter. Conversely, Harris et al. (2008), examined the effectiveness of SJ training with heavy loads (80%1RM) vs SJ at OPL loads (20–43.5%1RM) with elite rugby athletes showing that neither intervention was superiorly effective over the other for improving sprint ability even though the heavy load group became stronger. Albeit the OPL intervention group did exhibit lesser decrements in relative and absolute values of PP. However, it should be noted that a small portion of the training volume was allocated to the heavy load and OPL loads in this intervention (~20%), which may not have been sufficient in eliciting differential adaptive responses. This further highlights the importance of taking the training volume, competitive level, and initial strength levels into consideration when planning to use TDT over methods that use lighter loads.

In previous literature, relatively stronger individuals (i.e., squat 1RM $\sim 2.0 \text{ kg} \cdot \text{BM}^{-1}$) have responded better to more advanced training methods (e.g., CBT) (James et al., 2018; Newton & Kraemer, 1994). Considering training status is particularly important in terms of strength-power adaptations but also when interpreting meaningful changes in performance. For example, small increments in power performance (little as $\sim 1\%$) in highly trained individuals may be reported as non-statistically significant in training studies, but the ES may reflect a positive outcome which could deem a training method to be effective (Malcata & Hopkins, 2014; Rhea, 2004). Therefore, using TDT to increase baseline strength levels seems like a logical periodisation progression. Still, the overall greater time under tension (TUT), when using moderate to heavy loads at lower velocities in TDT may induce excessive muscle damage (Baumert et al., 2016; Wilson et al., 2012) and increase the overall metabolic stress placed on the neuromuscular system. This could be a disadvantage when used during congested periods of the season where speed-power qualities (i.e., vertical jump height, MPP, straight and curved sprinting, and COD ability) tend to be impaired (Freitas et al., 2021). Thus, perhaps TDT would be most beneficial when implemented in off-season periods rather than pre-season and in-season. In this context, a logical progression would be to shift to training methods that rely more on velocity-oriented lifts and movements that abide by the principles of velocity specificity (Cormie et al., 2011b) after the off-season, moving into preparatory periods like the pre-season. Therefore, practitioners should consider the pros and cons of TDT to avoid hindering the technical/tactical stimulus essential for achieving results in match fixtures (Berryman et al., 2018).

Combination training

The studies included reported trivial-very large effects on lower-body concentric MP and PP values supported by a high level of evidence in the literature. The CBT interventions in this review, consisted of using heavier-load exercises with lighter-load exercises within

the same session using various training methods paired together (e.g., traditional squats at 80–95%1RM with BM plyometric exercises). In previous literature, there is evidence that the use of CBT could potentially be superior to the lone use of other methods (e.g., TDT alone) (Spinetti et al., 2016, 2019) at improving lower-body power, despite disparities found with the acute and chronic use of CBT for PP improvements in differing populations (recreationally trained vs trained team-sport athletes) (Duthie et al., 2002; Talpey et al., 2014, 2016). This aligns with the wide range of effects observed in this review which could be explained by the heterogeneity displayed in the participant and intervention characteristics. Theoretically, greater enhancements in power using CBT would be logical, in the sense that it utilises various exercises that target specific areas of the force-velocity continuum (i.e., high-force, low-velocity to low-force, high-velocity) whereas using a training method in isolation may only target one area. The underlying physiological and performance mechanisms of adaptation with the use of CBT have been attributed to the targeting the above-mentioned areas of the force-velocity continuum within a session but also post-activation potentiation (PAP) (Seitz & Haff, 2016). However, the term ‘post-activation performance enhancement’ (PAPE) has been introduced in recent literature when referring to the enhancements in maximal strength, power, and speed following conditioning contractions (e.g., enhancement in sprint velocity following a set of back squats) whereas PAP should be referred to augmentation of muscular force/torque production during electrically evoked twitch following conditioning contractions (Blazevich & Babault, 2019; Boullosa et al., 2020; Prieske et al., 2020). Although, both PAPE and PAP can be present simultaneously following conditioning contractions (Zimmermann et al., 2021). Importantly, training status and relative strength may play a role in the adaptations following CBT. Stronger athletes have been reported to possess greater potentiation ability (i.e., the higher-load exercise may not excessively fatigue the athlete thereby creating an optimal condition to perform the subsequent lower-load power exercise) as well as an enhanced capability to produce mechanical power through distinctive force-time curve characteristics (e.g., net impulse forces during jumping) when compared to weaker individuals (Suchomel et al., 2016).

Recently, different systematic reviews with meta-analyses have assessed the effectiveness of CBT (Bauer et al., 2019; Marshall et al., 2021; Pagaduan et al., 2019). However, only few have examined the order in which the higher and lower load exercises were sequenced in a CBT session (e.g., complex vs contrast) specifically with team-sport athletes (Cormier et al., 2020; Freitas et al., 2017; Thapa et al., 2021). Overall, considerable positive effects have been found following complex training interventions (i.e., sequences alternating high-load with low-load, high-velocity exercises set-by-set) for adaptations in maximal strength, PP, jumping, sprinting, and COD ability with team-sport players (Cormier et al., 2020; Marshall et al., 2021). The potentially greater adaptations are theorised to be due to the presence of PAPE and PAP acutely in every session which then results in chronic adaptations after an accrued period of training. Further, these benefits have been found using CBT with higher level athletes with greater levels of strength in interventions >6 weeks in durations with moderate loads (<85%1RM) (Cormier et al., 2020). Therefore, based on the findings from this review, and previous literature, CBT is warranted for use in team-sport and may be a reasonable method to implement in various season periods due to the wide ranges of loads and intensities that can be manipulated within one session. Coaches and sport scientists may prefer this

method over others since it may result in the maintenance or improvement of strength (using the high-load exercises) whilst also developing the power/velocity (using low-load exercises) in every session.

Velocity-based training

VBT training was found to elicit positive effects on power development, with a moderate level of evidence on its effectiveness regarding concentric PP. In recent literature, VBT training has been discussed as a superior method over widely used percentage-based training to optimise individual programmes and identify an athlete's progression through an exercise programme, or as an accessory to percentage-based training (Guerriero et al., 2018). Mainly, velocities can be prescribed using mean velocity thresholds and/or velocity loss throughout a set with real-time visual (Randell et al., 2011) and/or verbal (Argus et al., 2011) feedback. This is said to increase competitiveness and motivation throughout the sets and repetitions of exercises (Weakley et al., 2019). Acutely, in one session, the use of VBT with 20% velocity loss thresholds can result in faster mean concentric and peak velocities across all repetitions compared to percentage-based training. Further, significantly less repetitions, TUT, and concentric TUT have been found when compared to percentage-based prescription (Banyard et al., 2019).

Chronically, superior improvements with VBT compared to percentage-based have been found with resistance-trained individuals (>2 years resistance training experience with a back squat 1RM to BM ratio of ~1.55 and above) (Banyard et al., 2020; Dorrell et al., 2020) in CMJ height, CMJ peak velocity, sprinting and COD ability after 6-week durations, using velocity zones (e.g., 0.74–0.88 m·s⁻¹ in the back squat) and velocity drops (i.e., 20% velocity loss threshold) to maintain intensity. Similarly, Orange et al. (2019) found greater velocity and power across sessions with lower TUT and perceptual training stress with academy rugby league players in a competitive season using VBT over percentage based training. This has important implications for coaches in team-sport settings where time can be limited, for-which the implementation of VBT could be a logical alternative. Using velocity to normalise the intensity at which force is applied at each concentric portion of a repetition may allow practitioners to adjust programmes based on adaptations that might have occurred on an in-between set and sessional basis whilst applying force with maximal intent in each repetition. However, it is important to note that every exercise has a load-velocity relationship that is task specific (González-Badillo & Sánchez-Medina, 2010; Pérez-Castilla et al., 2020), hence the minimum velocity thresholds that can be prescribed will be different depending upon the exercise (e.g., bench press vs back squat). Also, VBT is a novel autoregulation method that can be implemented in many ways thus recent reviews regarding VBT should be consulted for a summary of current practices (Balsalobre-Fernández & Torres-Ronda, 2021; Weakley et al., 2021). Further, results must be interpreted cautiously since there are limitations present and sometimes misapplication when using VBT with team-sport athletes thus the athlete's relative strength levels (strong vs weak), validity/reliability of instrumentation, and power calculation method (e.g., bar-power vs system-power) must be considered (Balsalobre-Fernández & Torres-Ronda, 2021; Jiménez-Reyes et al., 2021).

Furthermore, an argument concerning ecological validity could be made that the equipment used in VBT is not time efficient in situations where practitioners only have one device. On the other hand, in situations where multiple devices may be needed for time efficiency or in sports that have large rosters, the overall cost may be too great. Regardless, VBT as a method of prescribing exercise intensity and load is recommended for use with team-sport athletes to mitigate the variability in training adaptations due to individual strength-power characteristics. Further, VBT was shown to elicit performance improvements when applied in the off-season (Rauch et al., 2018), pre-season (Loturco et al., 2013), and in-season (Otero-Esquina et al., 2017) periods, indicating that this method may be beneficial in various periods of an annual periodisation with different types of team-sport players.

Optimal power load

Concerning OPL, this review assessed the results from a number of studies (Freitas et al., 2018; Loturco, Nakamura et al., 2016; Loturco et al., 2013; Rauch et al., 2018) reporting mainly trivial to large effect sizes; hence, the level of evidence supporting the use of this training method is high. OPL training within team-sports is often used due to its applicability to practice in congested periods and its ability to improve neuromuscular power at both ends of the force-velocity curve without the use of heavy loading methods (Freitas et al., 2018). Accordingly, this method can optimise power development in elite athletes concurrently with sport specific-training and matches possibly due to the maximal intent required to move the load as rapidly as possible for each repetition and in every session to produce mechanical power. Further, using OPL, similarly to VBT training, allows sport scientists and strength & conditioning coaches to manipulate the velocity and load used during strength-power sessions for each athlete, hence, individualising their training based on their needs. In alignment with this, Loturco et al. (2016) found that OPL resulted in large positive effects on BM MPP (ES = 1.7, 14.5%) and 40% BM MPP (ES = 1.2, 13.0%) compared to traditional strength-power periodisation using an HS and JS protocol within top-level soccer players (using a linear position transducer to measure MPP). Conversely, Freitas et al. (2018) reported trivial positive effects on CMJ PP (ES = 0.15, 2.9%) using OPL with an HS and hip thrust exercise protocol with semi-professional basketball players during a 6-week period (using force platforms to measure PP). Further, interestingly, Rauch et al. (2018) found that progressive VBT was equally as effective at improving concentric BS PP in comparison to OPL with volleyball players (ES: VBT = 0.86, 18.3% vs OPL = 0.79, 19.8% increase) (using a linear position transducer to measure PP). Notably, OPL can be described as a type of VBT training, since the velocity that maximises mechanical power is used (e.g., $1 \text{ m}\cdot\text{s}^{-1}$ in the half-squat), therefore it is no surprise that these methods of training elicit similar adaptations in power performance.

OPL training conditions may be better suited for use in pre-season (Loturco, Pereira et al., 2015) and in-season (Freitas et al., 2018; Loturco, Nakamura et al., 2016) periods to elicit improvements in high-velocity movements. Additionally, in many team-sports, recovery may be hindered for up to 72 h (Doeven et al., 2018) or more (McLean et al., 2010) following a match, and weekly concurrent technical-tactical and physical training volume may pose a problem for coaches wishing to maintain power- and speed-related

abilities, thus utilising methods such as OPL and VBT may be optimal for sessions with less TUT, shorter session times, whilst maximising power and velocity throughout.

Weightlifting and derivatives

Two WL studies (Helland et al., 2017; Hermassi et al., 2019) were included our analysis and trivial-moderate effects were found in concentric PP. Thus, the level of evidence in the current literature supporting the use of WL in team-sports to develop these specific power outcomes could be considered low. Still, it is important to note that WL movements are often incorporated alone or in combination with other methods (e.g., with TDT, plyometric or ballistic) into strength and conditioning programmes for athletes looking to improve their dynamic force production and can also be used as a new stimulus to strength training with athletes at a higher level when training becomes monotonous or new challenges need to be incorporated to optimise power development (i.e., training effect becomes stagnant meaning a new stimulus is required) (Haff & Nimphius, 2012). To improve various areas of an athlete's force-velocity (power) curve, many WL movements and/or their derivatives can be used to optimise areas where there are weaknesses by balancing the force or velocity components (James et al., 2020; Suchomel et al., 2017). Further, another aspect, and a benefit of WL, is that it can incorporate an important eccentric stimulus in the catch phase (i.e., load attenuation) of the lift which may not otherwise be achieved by other methods of training (Moolyk et al., 2013). Nevertheless, most coaches choose to include these types of exercises into their programmes since they involve triple extension actions (of the ankle, knee, and hip). In this context, WL pulling derivatives such as the clean pull, snatch pull, hang high pull, jump shrug, and mid-thigh pull may be good alternatives to full versions of WL exercises like cleans or power cleans and may produce a superior stimulus when compared to the full lifts (Comfort et al., 2011; DeWeese et al., 2013; Suchomel et al., 2015, 2020).

However, although effective, this type of training may be time consuming for practitioners to implement into their programmes as the exercises often require a greater amount of time dedicated to the technical execution of the lifts (Suchomel et al., 2015) to permit the athlete to develop adequate motor function to allow for proper loading. Given that WL elicits trivial-moderate effects and other methods in this review also elicited similar adaptations, some practitioners may opt to use less intricate training methods that require less coaching time. Accordingly, Helland et al. (2017) compared WL to CBT and MT and found that CBT and MT were more effective than WL at improving squat CMJ PP (measured using a linear position transducer) during an off-season. Interestingly, Hansen et al. (2011) combined traditional exercises (i.e., front and back squats) with WL (i.e., clean pull and power clean) in elite rugby union players during a pre-season period and showed positive improvements in PP at 40 kg load using traditional (i.e., 1% increase) and cluster (7.5%) loading schemes. These improvements were found in highly trained individuals; thus, there is evidence to support the use of TDT lifts alongside WL exercises (while minimising fatigue and maximising power/velocity using cluster loading). In fact, it is possible that such combination leads to superior improvements in power rather than using WL alone in team-sport athletes.

Eccentric overload training

Regarding EOT, our analysis was only performed on one study (Horwath et al., 2019); therefore, a low level of evidence persists. Nonetheless, the included investigation yielded moderate effects on power development in resistance-trained ice hockey players. JS MP increased by 12.5% (ES = 0.76) using EOT vs 17.5% (ES = 0.97) using CBT. It can be hypothesised that the utilisation of high-loads at low-velocities (e.g., EOT) as opposed to CBT methods that include exercises with lower-loads and higher-velocities could potentially explain the adaptations observed. Still, these can be mediated by the training background of individual athletes (i.e., familiarisation with eccentric-based training, resistance training experience and relative strength level). Despite this, EOT still resulted in MP enhancements possibly due to the variety of neural, neuromuscular, hormonal, and molecular adaptations attributed to eccentrically focused exercises (i.e., lengthening muscle actions) (Douglas et al., 2017). Specifically, production of higher peak force with lower levels of muscle activation, selective recruitment of higher-order motor units, greater cortical activity, stiffer muscle-tendon units, and lower metabolic costs (Douglas et al., 2017). Further, there are strong relationships between eccentric peak forces with CMJ PP and height in resistance trained individuals (Bridgeman et al., 2018). Altogether, these characteristics possibly contribute to an intra-session acute potentiation effect of EOT.

Regarding chronic strength-power adaptations following EOT, there appears to be a positive transfer to performance-based skills that are essential in multidirectional and intermittent-type sports (e.g., soccer, basketball, rugby, etc.) such as jumping (McNeill et al., 2019), sprinting (Krommes et al., 2017), and COD (Chaabene et al., 2018) ability. As far as implementing EOT in an annual training cycle, it would be rational to implement this method of training during the off-season or pre-season due to the importance of maintaining velocity-specific qualities in the competitive periods (i.e., in-season). However, a counter point to the previous may be related to the fact that in team-sports where maintaining strength within a season is crucial (e.g., football, ice hockey, and rugby), EOT can be effective at improving strength and maximal sprint velocity (Douglas et al., 2018). Thus, EOT may still be considered in competitive periods depending on the sport.

Flywheel training

Iso-inertial FW training possesses many similarities to EOT. The mechanisms of iso-inertial equipment (e.g., kBox, Versa Pulley, YoYo, etc.) allow for an overload stimulus throughout the concentric and eccentric phases of a lift (Maroto-Izquierdo et al., 2017). In this review, FW exhibited overall moderate-large effects on relative PP values; however, only one study was included resulting in a low level of evidence. Still, it is worth noting that this training intervention (Nuñez et al., 2018) concluded that using FW bilaterally was superior to unilateral (large and moderate effect, respectively) at enhancing relative PP with team-sport players. Further, long-term adaptations of performance-based actions have also been found using FW in strength training and rehabilitation programmes (De Hoyo et al., 2015, 2016). Due to the overloaded eccentric component of FW, neuromuscular characteristics are thought to be similar to that of EOT. Importantly, an aspect of FW and EOT worth

considering is the potentially elevated levels of muscle damage and soreness observed following the initial sessions. Yet, there seems to be a protective mechanism (García-López et al., 2007) against the occurrence of future injuries (De Hoyo et al., 2015) by incorporating an eccentric overload stimulus. Hence, when deciding the best seasonal period to implement FW and EOT, coaches should consider that the initial sessions may create adverse training effects (i.e., high magnitude of muscle damage). However, this is likely temporary and after growing accustomed to this methodology, players should be able to complete their training as planned. Also, another important aspect worth considering is that, in match-play, eccentric actions are not isolated nor performed at slow movement velocities. Rather, actions are performed in short time-periods, with most eccentric muscle actions occurring as part of the stretch-shortening cycle (i.e., storage of elastic energy to be subsequently used to increase the concentric action) or to rapidly decelerate the body (e.g., during COD manoeuvres). Therefore, movement specificity in the chosen training method must be taken into consideration in terms of the timing of implementation within an annual training plan. Exercises emphasising the eccentric portion of a movement using methods such as EOT and FW have been shown to improve many power values in team-sports, however, like in VBT and OPL, the equipment necessary may be too costly or time consuming to implement.

Mixed training

Some team-sport clubs have the ability to acquire the proper equipment for MT which allows VBT, OPL, FW, WL, plyometric, hill sprints, resisted sprints and other methods to be used interchangeably throughout the year. Coaches can combine these training methods in one session or over different sessions based on the physical demands of the season and the stimulus needed to maintain or improve performance-related outcomes. As previously mentioned, Otero-Esquina et al. (2017) used velocity zones between 1.28 and 1.07 m·s⁻¹ in the full squat exercise with professional soccer players in a training intervention that also incorporated FW, plyometric, and resisted sprint exercises and that resulted in meaningful improvements in relative CMJ PP. In this sense, the mechanisms of adaptation are thought to be similar to CBT. Specifically, the main adaptations may be related to the utilisation of exercises that target multiple points on the force-velocity curve from high-load low-velocity to low-load high-velocity and potentially from an accrue of potentiated (i.e., PAPE and PAP) strength-power training sessions over the intervention period. In theory, due to the multidirectional and intermittent nature of team-sports and the importance of developing qualities that are specific to actual match-play, it would be logical for coaches to prescribe MT interventions; still, more research is needed to confirm this assumption. MT is also characterised by relatively lighter loads moved at higher velocities which may be beneficial in crucial in-season periods where velocity-specific adaptations are key for success in competition. Finally, one challenging aspect with team-sports is prescribing exercise intensities for large rosters of players, thus an MT such as the protocol suggested by Otero-Esquina et al. (i.e., VBT exercises combined with plyometrics and sprints) may allow load individualisation for each athlete on the team whilst prescribing similar intensities based on the physical characteristics coaches wish to develop.

Limitations and future research

Some limitations can be found in this systematic review. First, there were a limited number of studies to be included in the VBT, OPL, WT, FW, EOT, and MT groups, thus findings must be interpreted cautiously. Secondly, the heterogeneity between participants, measurement and intervention characteristics could have affected our results. Moreover, in many studies, moments of the season were not reported, resistance training history was not clearly stated, and pre-post outcome data were missing. Such issues are problematic when assessing the effectiveness of the interventions on training outcomes. Also, no plyometric only studies met the inclusion criteria in this review, thus resulting in a lack of evidence concerning its effectiveness on the development of these specific power outcome measures (i.e., concentric MP, PP and MPP). However, it is important to note that plyometric training has been shown to be effective at improving other metrics of lower-body power in team-sports (Moran et al., 2020; Slimani et al., 2016). Further, there was a disproportionate number of studies conducted with males compared to females (Helland et al., 2017; Rauch et al., 2018) subjects, hence examining sex differences between the training methods listed in this review would be an interesting line of research. Finally, there was a lack of high-quality randomised control trials. Although the risk of bias was deemed 'low' with a mean PEDro score between 6 and 8, and study quality considered 'good', the methodological quality of studies could be improved in future research.

Importantly, we identified that most studies failed to adequately report the training loads and details about how exercises were performed. Mujika (2013) outlined two considerations that coaches and sport science researchers should take into account when quantifying training loads: (i) there are three distinct training loads: the planned loads before the season, loads prescribed on a daily basis, and the actual load completed by each individual athlete, and (ii) the actual load should be quantified, not only the prescribed load by the coach or researcher. This not only applies to the loads with strength and power training methods, but altogether with the technical-tactical sessions. Thus, the actual training loads (e.g., velocities, session ratings of perceived exertion, external workloads, etc.) of all concurrent training must be reported (if possible) in future studies related to training interventions in team-sport athletes. If actual training loads and intervention protocols are unclear, it is impossible to draw conclusions on the effectiveness of a training programme on power-related variables.

Practical applications

Based on the results from this systematic review, VBT, CBT, WL, OPL, TDT, FW, and EOT appear to be effective in improving concentric MP, PP and MPP outcomes. Importantly, no plyometric training only studies met the inclusion criteria which do not imply that this method is ineffective; rather, it simply means that no specific plyometric intervention studies have reported the abovementioned outcome variables with team-sport populations. The included methods have benefits on the development of lower-body absolute and relative power outcomes when used for durations between 6 to 12 weeks, 1 to 3 times per week with athletes 16 to 28 years of age from diverse types of team-sports even when performed concurrently with sport-specific training. Given the positive adaptations

obtained from many different training methods, coaches could benefit from utilising a variety of strategies throughout long and (sometimes) monotonous seasons, thereby seemingly increasing athlete buy-in regarding gym-based sessions. Further, when training effects become stagnant, various training methods can be manipulated to stimulate adaptations. Overall, the main recommendations following this review are: (i) team-sport athletes may benefit from developing appropriate levels of relative strength before progressing to more advanced training methods (i.e., WL, EOT, FW, CBT or MT) where TDT was shown to result in trivial to large ES in the current study; (ii) methods where barbell velocity or power can be monitored (e.g., OPL and VBT) on every repetition to maintain quality and intent are recommended in team-sports that train concurrently with high volumes of technical-technical training and congested schedules; and (iii) MT methods stimulating various areas of the force-velocity continuum (e.g., incorporating OPL half-squats, FW, plyometrics, and sprints used in the same session) may be beneficial. However, these should be prescribed by coaches and sport science practitioners taking into account actual (sport-specific) training loads.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Patrick Cormier  <http://orcid.org/0000-0001-6944-0467>

Tomás T. Freitas  <http://orcid.org/0000-0001-8571-3189>

References

- Aagaard, P. (2003). Training-induced changes in neural function. *Exercise and Sport Sciences Reviews*, 31(2), 61–67. <https://doi.org/10.1097/00003677-200304000-00002>
- Argus, C. K., Gill, N. D., Keogh, J. W., & Hopkins, W. G. (2011). Acute effects of verbal feedback on upper-body performance in elite athletes. *Journal of Strength and Conditioning Research*, 25(12), 3282–3287. <https://doi.org/10.1519/JSC.0b013e3182133b8c>
- Balsalobre-Fernández, C., & Torres-Ronda, L. (2021). The implementation of velocity-based training paradigm for team sports: Framework, technologies, practical recommendations and challenges. *Sports*, 9(4), 47. <https://doi.org/10.3390/sports9040047>
- Banyard, H. G., Tufano, J. J., Delgado, J., Thompson, S. W., & Nosaka, K. (2019). Comparison of the effects of velocity-based training methods and traditional 1RM-percent-based training prescription on acute kinetic and kinematic variables. *International Journal of Sports Physiology and Performance*, 14(2), 246–255. <https://doi.org/10.1123/ijsp.2018-0147>
- Banyard, H. G., Tufano, J. J., Weakley, J. J. S., Wu, S., Jukic, I., & Nosaka, K. (2020). Superior changes in jump, sprint, and change-of-direction performance but not maximal strength following 6 weeks of velocity-based training compared with 1-repetition-maximum percentage-based training. *International Journal of Sports Physiology and Performance*, 16(2), 232–34. <https://doi.org/10.1123/ijsp.2019-0999>
- Bauer, P., Uebellacker, F., Mitter, B., Aigner, A. J., Hasenoehrl, T., Ristl, R., Tschan, H., & Seitz, L. B. (2019). Combining higher-load and lower-load resistance training exercises: A systematic review and meta-analysis of findings from complex training studies. *Journal of Science & Medicine in Sport*, 22(7), 838–851. <https://doi.org/10.1016/j.jsams.2019.01.006>

- Baumert, P., Lake, M. J., Stewart, C. E., Drust, B., & Erskine, R. M. (2016). Genetic variation and exercise-induced muscle damage: Implications for athletic performance, injury and ageing. *European Journal of Applied Physiology*, 116, 1595–1625. <https://doi.org/10.1007/s00421-016-3411-1>
- Berryman, N., Mujika, I., & Bosquet, L. (2018). Concurrent training for sports performance: The two sides of the medal. *International Journal of Sports Physiology and Performance*, 14(3), 279–285. <https://doi.org/10.1123/ijsp.2018-0103>
- Blazevich, A. J., & Babault, N. (2019). Post-activation potentiation versus post-activation performance enhancement in humans: Historical perspective, underlying mechanisms, and current issues. *Frontiers in Physiology*, 10, 1359. <https://doi.org/10.3389/fphys.2019.01359>
- Boullosa, D., Beato, M., Dello Iacono, A., Cuenca-Fernández, F., Doma, K., Schumann, M., Zagatto, A. M., Loturco, I., & Behm, D. G. (2020). A new taxonomy for postactivation potentiation in sport. *International Journal of Sports Physiology and Performance*, 15(1), 1197–1200. <https://doi.org/10.1123/ijsp.2020-0350>
- Bridgeman, L. A., McGuigan, M. R., Gill, N. D., & Dulson, D. K. (2018). Relationships between concentric and eccentric strength and countermovement jump performance in resistance trained men. *Journal of Strength and Conditioning Research*, 32(1), 255–260. <https://doi.org/10.1519/JSC.0000000000001539>
- Cashin, A. G., & McAuley, J. H. (2020). Clinimetrics: Physiotherapy evidence database (PEDro) scale. *Journal of Physiotherapy*, 66(1), 59. <https://doi.org/10.1016/j.jphys.2019.08.005>
- Chaabene, H., Negra, Y., Moran, J., Prieske, O., Sammoud, S., Ramirez-Campillo, R., & Granacher, U. (2019). Plyometric training improves not only measures of linear speed, power, and change-of-direction speed but also repeated sprint ability in female young handball players. *Journal of Strength and Conditioning Research*, Epub Ahead of Print. <https://doi.org/10.1519/JSC.0000000000003128>
- Chaabene, H., Prieske, O., Negra, Y., & Granacher, U. (2018). Change of direction speed: Toward a strength training approach with accentuated eccentric muscle actions. *Sports Medicine (Auckland, N.Z.)*, 48(8), 1773–1779. <https://doi.org/10.1007/s40279-018-0907-3>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences*. Routledge. <https://doi.org/10.4324/9780203771587>
- Comfort, P., Allen, M., & Graham-Smith, P. (2011). Comparisons of peak ground reaction force and rate of force development during variations of the power clean. *The Journal of Strength & Conditioning Research*, 25(5), 1235–1239. <https://doi.org/10.1519/JSC.0b013e3181d6dc0d>
- Cook, C. J., Beaven, C. M., & Kilduff, L. P. (2013). Three weeks of eccentric training combined with overspeed exercises enhances power and running speed performance gains in trained athletes. *Journal of Strength and Conditioning Research*, 27(5), 1280–1286. <https://doi.org/10.1519/JSC.0b013e3182679278>
- Cormie, P., McBride, J. M., & McCaulley, G. O. (2007). Validation of power measurement techniques in dynamic lower body resistance exercises. *Journal of Applied Biomechanics*, 23(2), 103–118. <https://doi.org/10.1123/jab.23.2.103>
- Cormie, P., McBride, J. M., & McCaulley, G. O. (2009). Power-time, force-time, and velocity-time curve analysis of the countermovement jump: Impact of training. *Journal of Strength and Conditioning Research*, 23(1), 177–186. <https://doi.org/10.1519/JSC.0b013e3181889324>
- Cormie, P., McGuigan, M. R., & Newton, R. U. (2011a). Developing maximal neuromuscular power: Part 1--biological basis of maximal power production. *Sports Medicine (Auckland, N.Z.)*, 41(1), 17–38. <https://doi.org/10.2165/11537690-000000000-00000>
- Cormie, P., McGuigan, M. R., & Newton, R. U. (2011b). Developing maximal neuromuscular power: Part 2 - training considerations for improving maximal power production. *Sports Medicine (Auckland, N.Z.)*, 41(2), 125–146. <https://doi.org/10.2165/11538500-000000000-00000>
- Cormier, P., Freitas, T. T., Rubio-Arias, J. A., & Alcaraz, P. E. (2020). Complex and contrast training: Does strength and power training sequence affect performance-based adaptations in team sports? A systematic review and meta-analysis. *Journal of Strength and Conditioning Research*, 34(5), 1461–1479. <https://doi.org/10.1519/JSC.0000000000003493>

- Cronin, J., & Sleivert, G. (2005). Challenges in understanding the influence of maximal power training on improving athletic performance. *Sports Medicine*, 35(3), 213–234. <https://doi.org/10.2165/00007256-200535030-00003>
- de Hoyo, M., Pozzo, M., Sañudo, B., Carrasco, L., Gonzalo-Skok, O., Domínguez-Cobo, S., & Morán-Camacho, E. (2015). Effects of a 10-week in-season eccentric-overload training program on muscle-injury prevention and performance in junior elite soccer players. *International Journal of Sports Physiology and Performance*, 10(1), 46–52. <https://doi.org/10.1123/ijsspp.2013-0547>
- de Hoyo, M., Sañudo, B., Carrasco, L., Mateo-Cortes, J., Domínguez-Cobo, S., Fernandes, O., Del Ojo, J. J., & Gonzalo-Skok, O. (2016). Effects of 10-week eccentric overload training on kinetic parameters during change of direction in football players. *Journal of Sports Sciences*, 34(14), 1380–1387. <https://doi.org/10.1080/02640414.2016.1157624>
- DeWeese, B. H., Serrano, A. J., Scruggs, S. K., & Burton, J. D. (2013). The midhigh pull: Proper application and progressions of a weightlifting movement derivative. *Strength & Conditioning Journal*, 35(6), 54–58. <https://doi.org/10.1519/SSC.0b013e318297c77b>
- Doeven, S. H., Brink, M. S., Kosse, S. J., & Lemmink, K. A. P. M. (2018). Postmatch recovery of physical performance and biochemical markers in team ball sports: A systematic review. *BMJ Open Sport & Exercise Medicine*, 4(1), e000264. <https://doi.org/10.1136/bmjsem-2017-000264>
- Dorrell, H. F., Smith, M. F., & Gee, T. I. (2020). Comparison of velocity-based and traditional percentage-based loading methods on maximal strength and power adaptations. *The Journal of Strength & Conditioning Research*, 34(1), 46–53. <https://doi.org/10.1519/JSC.0000000000003089>
- Dos'Santos, T., Thomas, C., Jones, P. A., & Comfort, P. (2017). Mechanical determinants of faster change of direction speed performance in male athletes. *Journal of Strength and Conditioning Research*, 31(3), 696–705. <https://doi.org/10.1519/JSC.0000000000001535>
- Douglas, J., Pearson, S., Ross, A., & McGuigan, M. (2017). Eccentric exercise: Physiological characteristics and acute responses. *Sports Medicine (Auckland, N.Z.)*, 47(4), 663–675. <https://doi.org/10.1007/s40279-016-0624-8>
- Douglas, J., Pearson, S., Ross, A., & McGuigan, M. (2018). Effects of accentuated eccentric loading on muscle properties, strength, power, and speed in resistance-trained rugby players. *Journal of Strength and Conditioning Research*, 32(10), 2750–2761. <https://doi.org/10.1519/JSC.0000000000002772>
- Dugan, E. L., Doyle, T. L. A., Humphries, B., Hasson, C. J., & Newton, R. U. (2004). Determining the optimal load for jump squats: A review of methods and calculations. *Journal of Strength and Conditioning Research*, 18(3), 668–674.
- Duthie, G. M., Young, W. B., & Aitken, D. A. (2002). The acute effects of heavy loads on jump squat performance: An evaluation of the complex and contrast methods of power development. *Journal of Strength and Conditioning Research*, 16(4), 530–538.
- Freitas, T. T., Calleja-González, J., Carlos-Vivas, J., Marín-Cascales, E., & Alcaraz, P. E. (2018). Short-term optimal load training vs a modified complex training in semi-professional basketball players. *Journal of Sports Sciences* 37(4), 434–442. <https://doi.org/10.1080/02640414.2018.1504618>
- Freitas, T. T., Martinez-Rodriguez, A., Calleja-González, J., & Alcaraz, P. E. (2017). Short-term adaptations following Complex Training in team-sports: A meta-analysis. *PloS One*, 12(6), e0180223. <https://doi.org/10.1371/journal.pone.0180223>
- Freitas, T. T., Pereira, L. A., Reis, V. P., Fernandes, V., Alcaraz, P. E., Azevedo, P. H. S. M., & Loturco, I. (2021). Effects of a congested fixture period on speed and power performance of elite young soccer players. *International Journal of Sports Physiology and Performance*, 16(8), 1120–1126. <https://doi.org/10.1123/ijsspp.2020-0280>
- García-López, D., Cuevas, M. J., Almar, M., Lima, E., De Paz, J. A., & González-Gallego, J. (2007). Effects of eccentric exercise on NF-kappaB activation in blood mononuclear cells. *Medicine and Science in Sports and Exercise*, 39(4), 653–664. <https://doi.org/10.1249/mss.0b013e31802f04f6>
- González-Badillo, J. J., & Sánchez-Medina, L. (2010). Movement velocity as a measure of loading intensity in resistance training. *International Journal of Sports Medicine*, 31(5), 347–352. <https://doi.org/10.1055/s-0030-1248333>

- Guerriero, A., Varalda, C., & Piacentini, M. F. (2018). The role of velocity-based training in the strength periodization for modern athletes. *Journal of Functional Morphology and Kinesiology*, 3(4), 55. <https://doi.org/10.3390/jfmk3040055>
- Haff, G. G., & Nimphius, S. (2012). Training principles for power. *Strength & Conditioning Journal*, 34(6), 2–12. <https://doi.org/10.1519/SSC.0b013e31826db467>
- Häkkinen, K. (1989). Neuromuscular and hormonal adaptations during strength and power training. A review. *The Journal of Sports Medicine and Physical Fitness*, 29(1), 9–26.
- Hammami, M., Negra, Y., Billaut, F., Hermassi, S., Shephard, R. J., & Chelly, M. S. (2018). Effects of lower-limb strength training on agility, repeated sprinting with changes of direction, leg peak power, and neuromuscular adaptations. *Journal of Strength and Conditioning Research*, 32(1), 37–47. <https://doi.org/10.1519/JSC.0000000000001813>
- Hansen, K. T., Cronin, J. B., & Newton, M. J. (2011). The effect of cluster loading on force, velocity, and power during ballistic jump squat training. *International Journal of Sports Physiology & Performance*, 6(4), 455–468. <https://doi.org/10.1123/ijsp.6.4.455>
- Harper, D. J., Carling, C., & Kiely, J. (2019). High-intensity acceleration and deceleration demands in elite team sports competitive match play: A systematic review and meta-analysis of observational studies. *Sports Medicine*, 49(12), 1923–1947. <https://doi.org/10.1007/s40279-019-01170-1>
- Harris, N. K., Cronin, J., Taylor, K.-L., Boris, J., & Sheppard, J. (2010). Understanding position transducer technology for strength and conditioning practitioners. *Strength & Conditioning Journal*, 32(4), 66–79. <https://doi.org/10.1519/SSC.0b013e3181eb341b>
- Harris, N. K., Cronin, J. B., Hopkins, W. G., & Hansen, K. T. (2008). Squat jump training at maximal power loads vs. heavy loads: Effect on sprint ability. *Journal of Strength and Conditioning Research*, 22(6), 1742–1749. <https://doi.org/10.1519/JSC.0b013e318187458a>
- Helland, C., Hole, E., Iversen, E., Olsson, M. C., Seynnes, O., Solberg, P. A., & Paulsen, G. (2017). Training strategies to improve muscle power: Is Olympic-style weightlifting relevant? *Medicine and Science in Sports and Exercise*, 49(4), 736–745. <https://doi.org/10.1249/MSS.0000000000001145>
- Hermassi, S., Chelly, M. S., Tabka, Z., Shephard, R. J., & Chamari, K. (2011). Effects of 8-week in-season upper and lower limb heavy resistance training on the peak power, throwing velocity, and sprint performance of elite male handball players. *Journal of Strength and Conditioning Research*, 25(9), 2424–2433. <https://doi.org/10.1519/JSC.0b013e3182030edb>
- Hermassi, S., Schwesig, R., Aloui, G., Shephard, R. J., & Chelly, M. S. (2019). Effects of short-term in-season weightlifting training on the muscle strength, peak power, sprint performance, and ball-throwing velocity of male handball players. *Journal of Strength and Conditioning Research*, 33(12), 3309–3321. <https://doi.org/10.1519/JSC.0000000000003068>
- Hopkins, W. G., Marshall, S. W., Batterham, A. M., & Hanin, J. (2009). Progressive statistics for studies in sports medicine and exercise science. *Medicine and Science in Sports and Exercise*, 41(1), 3–13. <https://doi.org/10.1249/MSS.0b013e31818cb278>
- Horwath, O., Paulsen, G., Esping, T., Seynnes, O., & Olsson, M. C. (2019). Isokinetic resistance training combined with eccentric overload improves athletic performance and induces muscle hypertrophy in young ice hockey players. *Journal of Science & Medicine in Sport*, 22(7), 821–826. <https://doi.org/10.1016/j.jsams.2018.12.017>
- Ishida, A., Rochau, K., Findlay, K. P., Devero, B., Duca, M., & Stone, M. H. (2020). Effects of an initial muscle strength level on sports performance changes in collegiate soccer players. *Sports*, 8(9), 127. <https://doi.org/10.3390/sports8090127>
- James, L. P., Gregory Haff, G., Kelly, V. G., Connick, M. J., Hoffman, B. W., & Beckman, E. M. (2018). The impact of strength level on adaptations to combined weightlifting, plyometric, and ballistic training. *Scandinavian Journal of Medicine & Science in Sports*, 28(5), 1494–1505. <https://doi.org/10.1111/sms.13045>
- James, L. P., Suchomel, T. J., Comfort, P., Haff, G. G., & Connick, M. J. (2020). Rate of force development adaptations after weightlifting-style training: The influence of power clean ability. *The Journal of Strength & Conditioning Research*, Publish Ahead of Print. <https://doi.org/10.1519/JSC.00000000000003673>

- Jiménez-Reyes, P., Castaño-Zambudio, A., Cuadrado-Peñafiel, V., González-Hernández, J. M., Capelo-Ramírez, F., Martínez-Aranda, L. M., & González-Badillo, J. J. (2021). Differences between adjusted vs. non-adjusted loads in velocity-based training: Consequences for strength training control and programming. *Peer Journal*, 9, e10942. <https://doi.org/10.7717/peerj.10942>
- Kawamori, N., & Haff, G. G. (2004). The optimal training load for the development of muscular power. *The Journal of Strength & Conditioning Research*, 18(3), 675.
- Kawamori, N., Nosaka, K., & Newton, R. U. (2013). Relationships between ground reaction impulse and sprint acceleration performance in team sport athletes. *Journal of Strength and Conditioning Research*, 27(3), 568–573. <https://doi.org/10.1519/JSC.0b013e318257805a>
- Kirby, T. J., McBride, J. M., Haines, T. L., & Dayne, A. M. (2011). Relative net vertical impulse determines jumping performance. *Journal of Applied Biomechanics*, 27(3), 207–214. <https://doi.org/10.1123/jab.27.3.207>
- Krommes, K., Petersen, J., Nielsen, M. B., Aagaard, P., Hölmich, P., & Thorborg, K. (2017). Sprint and jump performance in elite male soccer players following a 10-week Nordic Hamstring exercise Protocol: A randomised pilot study. *BMC Research Notes*, 10(1), 669. <https://doi.org/10.1186/s13104-017-2986-x>
- Lake, J. P., Lauder, M. A., & Smith, N. A. (2012). Barbell kinematics should not be used to estimate power output applied to the Barbell-and-body system center of mass during lower-body resistance exercise. *Journal of Strength and Conditioning Research*, 26(5), 1302–1307. <https://doi.org/10.1519/JSC.0b013e31822e7b48>
- Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. *Frontiers in Psychology*, 4, article 863. <https://doi.org/10.3389/fpsyg.2013.00863>
- Lauersen, J. B., Andersen, T. E., & Andersen, L. B. (2018). Strength training as superior, dose-dependent and safe prevention of acute and overuse sports injuries: A systematic review, qualitative analysis and meta-analysis. *British Journal of Sports Medicine*, 52(24), 1557–1563. <https://doi.org/10.1136/bjsports-2018-099078>
- Los Arcos, A., Yanci, J., Mendiguchia, J., Salinero, J. J., Brughelli, M., & Castagna, C. (2014). Short-term training effects of vertically and horizontally oriented exercises on neuromuscular performance in professional soccer players. *International Journal of Sports Physiology and Performance*, 9(3), 480–488. <https://doi.org/10.1123/ijspp.2013-0063>
- Loturco, I., Bishop, C., Freitas, T. T., Pereira, L. A., & Jeffreys, I. (2020). Vertical force production in soccer: Mechanical aspects and applied training strategies. *Strength & Conditioning Journal*, 42(2), 6–15. <https://doi.org/10.1519/SSC.0000000000000513>
- Loturco, I., Nakamura, F. Y., Kobal, R., Gil, S., Pivetti, B., Pereira, L. A., & Roschel, H. (2016). Traditional periodization versus optimum training load applied to soccer players: Effects on neuromuscular abilities. *International Journal of Sports Medicine*, 37(13), 1051–1059. <https://doi.org/10.1055/s-0042-107249>
- Loturco, I., Nakamura, F. Y., Tricoli, V., Kobal, R., Cal Abad, C. C., Kitamura, K., Ugrinowitsch, C., Gil, S., Pereira, L. A., & Gonzalez-Badillo, J. J. (2015). Determining the optimum power load in jump squat using the mean propulsive velocity. *PloS One*, 10(10), e0140102. <https://doi.org/10.1371/journal.pone.0140102>
- Loturco, I., Pereira, L. A., Kobal, R., Kitamura, K., Cal Abad, C. C., Nakamura, F. Y., & Pai, C. N. (2017). Peak versus mean propulsive power outputs: Which is more closely related to jump squat performance? *The Journal of Sports Medicine and Physical Fitness*, 57(11), 1432–1444. <https://doi.org/10.23736/S0022-4707.17.06940-7>
- Loturco, I., Pereira, L. A., Kobal, R., Zanetti, V., Gil, S., Kitamura, K., Abad, C. C. C., & Nakamura, F. Y. (2015). Half-squat or jump squat training under optimum power load conditions to counteract power and speed decrements in Brazilian elite soccer players during the preseason. *Journal of Sports Sciences*, 33(12), 1283–1292. <https://doi.org/10.1080/02640414.2015.1022574>
- Loturco, I., Pereira, L. A., Zanetti, V., Kitamura, K., Abad, C. C. C., Kobal, R., & Nakamura, F. Y. (2016). Mechanical differences between barbell and body optimum power loads in the jump squat exercise. *Journal of Human Kinetics*, 54(1), 153–162. <https://doi.org/10.1515/hukin-2016-0044>

- Loturco, I., Suchomel, T., Bishop, C., Kobal, R., Pereira, L. A., & McGuigan, M. (2018). 1RM measures or maximum bar-power output: Which is more related to sport performance? *International Journal of Sports Physiology and Performance*, 14(1), 33–37. <https://doi.org/10.1123/ijsp.2018-0255>
- Loturco, I., Suchomel, T., James, L. P., Bishop, C., Abad, C. C. C., Pereira, L. A., & McGuigan, M. R. (2018). Selective influences of maximum dynamic strength and bar-power output on team sports performance: A comprehensive study of four different disciplines. *Frontiers in Physiology*, 9, 1820. <https://doi.org/10.3389/fphys.2018.01820>
- Loturco, I., Ugrinowitsch, C., Tricoli, V., Pivetti, B., & Roschel, H. (2013). Different loading schemes in power training during the preseason promote similar performance improvements in Brazilian elite soccer players. *Journal of Strength & Conditioning Research*, 27(7), 1791–1797. <https://doi.org/10.1519/JSC.0b013e3182772da6>
- Maher, C. G., Sherrington, C., Herbert, R. D., Moseley, A. M., & Elkins, M. (2003). Reliability of the PEDro scale for rating quality of randomized controlled trials. *Physical Therapy*, 83(8), 713–721. <https://doi.org/10.1093/ptj/83.8.713>
- Malcata, R. M., & Hopkins, W. G. (2014). Variability of competitive performance of elite athletes: A systematic review. *Sports Medicine (Auckland, N.Z.)*, 44(12), 1763–1774. <https://doi.org/10.1007/s40279-014-0239-x>
- Malone, S., Hughes, B., Doran, D. A., Collins, K., & Gabbett, T. J. (2019). Can the workload-injury relationship be moderated by improved strength, speed and repeated-sprint qualities? *Journal of Science and Medicine in Sport*, 22(1), 29–34. <https://doi.org/10.1016/j.jsams.2018.01.010>
- Mann, J. B., Ivey, P. A., & Sayers, S. P. (2015). Velocity-based training in football. *Strength & Conditioning Journal*, 37(6), 52–57. <https://doi.org/10.1519/SSC.0000000000000177>
- Maroto-Izquierdo, S., Fernandez-Gonzalo, R., Magdi, H. R., Manzano-Rodriguez, S., González-Gallego, J., & De Paz, J. A. (2019). Comparison of the musculoskeletal effects of different iso-inertial resistance training modalities: Flywheel vs. Electric-motor. *European Journal of Sport Science*, 19(9), 1184–1194. <https://doi.org/10.1080/17461391.2019.1588920>
- Maroto-Izquierdo, S., García-López, D., Fernandez-Gonzalo, R., Moreira, O. C., González-Gallego, J., & de Paz, J. A. (2017). Skeletal muscle functional and structural adaptations after eccentric overload flywheel resistance training: A systematic review and meta-analysis. *Journal of Science and Medicine in Sport*, 20(10), 943–951. <https://doi.org/10.1016/j.jsams.2017.03.004>
- Marshall, J., Bishop, C., Turner, A., & Haff, G. G. (2021). Optimal training sequences to develop lower body force, velocity, power, and jump height: A systematic review with meta-analysis. *Sports Medicine*, 51(6), 1245–1271. <https://doi.org/10.1007/s40279-021-01430-z>
- McBride, J. M., Haines, T. L., & Kirby, T. J. (2011). Effect of loading on peak power of the bar, body, and system during power cleans, squats, and jump squats. *Journal of Sports Sciences*, 29(11), 1215–1221. <https://doi.org/10.1080/02640414.2011.587444>
- McBride, J. M., Kirby, T. J., Haines, T. L., & Skinner, J. (2010). Relationship between relative net vertical impulse and jump height in jump squats performed to various squat depths and with various loads. *International Journal of Sports Physiology and Performance*, 5(4), 484–496. <https://doi.org/10.1123/ijsp.5.4.484>
- McGuigan, M. R., Wright, G. A., & Fleck, S. J. (2012). Strength training for athletes: Does it really help sports performance? *International Journal of Sports Physiology and Performance*, 7(1), 2–5. <https://doi.org/10.1123/ijsp.7.1.2>
- McLean, B. D., Coutts, A. J., Kelly, V., McGuigan, M. R., & Cormack, S. J. (2010). Neuromuscular, endocrine, and perceptual fatigue responses during different length between-match microcycles in professional rugby league players. *International Journal of Sports Physiology and Performance*, 5(3), 367–383. <https://doi.org/10.1123/ijsp.5.3.367>
- McMaster, D. T., Gill, N., Cronin, J., & McGuigan, M. (2014). A brief review of strength and ballistic assessment methodologies in sport. *Sports Medicine (Auckland, N.Z.)*, 44(5), 603–623. <https://doi.org/10.1007/s40279-014-0145-2>
- McNeill, C., Beaven, C. M., McMaster, D. T., & Gill, N. (2019). Eccentric training interventions and team sport athletes. *Journal of Functional Morphology and Kinesiology*, 4(4), 67. <https://doi.org/10.3390/jfmk4040067>

- Meylan, C. M. P., Cronin, J. B., Oliver, J. L., Hughes, M. G., & Manson, S. (2014). An evidence-based model of power development in youth soccer. *International Journal of Sports Science & Coaching*, 9(5), 1241–1264. <https://doi.org/10.1260/1747-9541.9.5.1241>
- Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G., & Group, T. P. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *PLOS Medicine*, 6(7), e1000097. <https://doi.org/10.1371/journal.pmed.1000097>
- Moolyk, A. N., Carey, J. P., & Chiu, L. Z. F. (2013). Characteristics of lower extremity work during the impact phase of jumping and weightlifting. *Journal of Strength and Conditioning Research*, 27(12), 3225–3232. <https://doi.org/10.1519/JSC.0b013e31828ddff9>
- Moran, J., Ramirez-Campillo, R., Liew, B., Chaabene, H., Behm, D. G., García-Hermoso, A., Izquierdo, M., & Granacher, U. (2020). Effects of vertically and horizontally orientated plyometric training on physical performance: A meta-analytical comparison. *Sports Medicine*, 51(1), 65–79. <https://doi.org/10.1007/s40279-020-01340-6>
- Mujika, I. (2013). The alphabet of sport science research starts with Q. *International Journal of Sports Physiology and Performance*, 8(5), 465–466. <https://doi.org/10.1123/ijsp.8.5.465>
- Newton, R. U., & Kraemer, W. J. (1994). Developing explosive muscular power: Implications for a mixed methods training strategy. *Strength & Conditioning Journal*, 16(5), 20.
- Núñez, F. J., Santalla, A., Carrasquilla, I., Asian, J. A., Reina, J. I., & Suarez-Arrones, L. J. (2018). The effects of unilateral and bilateral eccentric overload training on hypertrophy, muscle power and COD performance, and its determinants, in team sport players. *Plos One*, 13(3), e0193841. <https://doi.org/10.1371/journal.pone.0193841>
- Orange, S. T., Metcalfe, J. W., Robinson, A., Applegarth, M. J., & Liefieith, A. (2019). Effects of in-season velocity- versus percentage-based training in academy rugby league players. *International Journal of Sports Physiology and Performance*, 15(4), 554–561. <https://doi.org/10.1123/ijsp.2019-0058>
- Otero-Esquina, C., de Hoyo Lora, M., Gonzalo-Skok, Ó., Domínguez-Cobo, S., & Sánchez, H. (2017). Is strength-training frequency a key factor to develop performance adaptations in young elite soccer players? *European Journal of Sport Science*, 17(10), 1241–1251. <https://doi.org/10.1080/17461391.2017.1378372>
- Pagaduan, J., Schoenfeld, B., & Pojskic, H. (2019). Systematic review and meta-analysis on the effect of contrast training on vertical jump performance. *Strength & Conditioning Journal*, 41(3), 63–78. <https://doi.org/10.1519/SSC.0000000000000442>
- Pérez-Castilla, A., García-Ramos, A., Padial, P., Morales-Artacho, A. J., & Feriche, B. (2020). Load-velocity relationship in variations of the half-squat exercise: Influence of execution technique. *Journal of Strength and Conditioning Research*, 34(4), 1024–1031. <https://doi.org/10.1519/JSC.0000000000002072>
- Prieske, O., Behrens, M., Chaabene, H., Granacher, U., & Maffiuletti, N. A. (2020). Time to differentiate postactivation “potentiation” from “performance enhancement” in the strength and conditioning community. *Sports Medicine*, 50(9), 1559–1565. <https://doi.org/10.1007/s40279-020-01300-0>
- Ramirez, J. M., Nunez, V. M., Lancho, C., Poblador, M. S., & Lancho, J. L. (2015). Velocity-based training of lower limb to improve absolute and relative power outputs in concentric phase of half-squat in soccer players. *Journal of Strength and Conditioning Research*, 29(11), 3084–3088. <https://doi.org/10.1519/JSC.0000000000000407>
- Randell, A. D., Cronin, J. B., Keogh, J. W. L., Gill, N. D., & Pedersen, M. C. (2011). Effect of instantaneous performance feedback during 6 weeks of velocity-based resistance training on sport-specific performance tests. *Journal of Strength and Conditioning Research*, 25(1), 87–93. <https://doi.org/10.1519/JSC.0b013e3181fee634>
- Rauch, J. T., Loturco, I., Cheesman, N., Thiel, J., Alvarez, M., Miller, N., Carpenter, N., Barakat, C., Velasquez, G., Stanjones, A., Aube, D., Andersen, J. C., & De Souza, E. O. (2018). Similar strength and power adaptations between two different velocity-based training regimens in collegiate female volleyball players. *Sports (Basel, Switzerland)*, 6(4), 163. <https://doi.org/10.3390/sports6040163>

- Rhea, M. R. (2004). Determining the magnitude of treatment effects in strength training research through the use of the effect size. *Journal of Strength and Conditioning Research*, 18(4), 918–920. <https://doi.org/10.1519/14403.1>
- Ribeiro, J., Teixeira, L., Lemos, R., Teixeira, A. S., Moreira, V., Silva, P., & Nakamura, F. Y. (2020). Effects of plyometric versus optimum power load training on components of physical fitness in young male soccer players. *International Journal of Sports Physiology and Performance*, 15(2), 222–230. <https://doi.org/10.1123/ijsspp.2019-0039>
- Rønnestad, B. R., Kvamme, N. H., Sundé, A., & Raastad, T. (2008). Short-term effects of strength and plyometric training on sprint and jump performance in professional soccer players. *Journal of Strength and Conditioning Research*, 22(3), 773–780. <https://doi.org/10.1519/JSC.0b013e31816a5e86>
- Sanchez-Medina, L., Perez, C. E., & Gonzalez-Badillo, J. J. (2010). Importance of the propulsive phase in strength assessment. *International Journal of Sports Medicine*, 31(2), 123–129. <https://doi.org/10.1055/s-0029-1242815>
- Sapega, A. A., & Drillings, G. (1983). The definition and assessment of muscular power. *The Journal of Orthopaedic and Sports Physical Therapy*, 5(1), 7–9. <https://doi.org/10.2519/jospt.1983.5.1.7>
- Seitz, L. B., & Haff, G. G. (2016). Factors modulating post-activation potentiation of jump, sprint, throw, and upper-body ballistic performances: A systematic review with meta-analysis. *Sports Medicine (Auckland, N.Z.)*, 46(2), 231–240. <https://doi.org/10.1007/s40279-015-0415-7>
- Slimani, M., Chamari, K., Miarka, B., Del Vecchio, F. B., & Cheour, F. (2016). Effects of plyometric training on physical fitness in team sport athletes: A systematic review. *Journal of Human Kinetics*, 53(1), 231–247. <https://doi.org/10.1515/hukin-2016-0026>
- Soriano, M. A., Jiménez-Reyes, P., Rhea, M. R., & Marín, P. J. (2015). The optimal load for maximal power production during lower-body resistance exercises: A meta-analysis. *Sports Medicine (Auckland, N.Z.)*, 45(8), 1191–1205. <https://doi.org/10.1007/s40279-015-0341-8>
- Soriano, M. A., Kipp, K., Lake, J. P., Suchomel, T. J., Marín, P. J., Sainz De Baranda, M. P., & Comfort, P. (2020). Mechanical power production assessment during weightlifting exercises. A systematic review. *Sports Biomechanics*, 1752–6116. <https://doi.org/10.1080/14763141.2020.1747529>
- Spinetti, J., Figueiredo, T., Bastos De Oliveira, V., Assis, M., Fernandes De Oliveira, L., Miranda, H., Machado De Ribeiro Reis, V. M., & Simao, R. (2016). Comparison between traditional strength training and complex contrast training on repeated sprint ability and muscle architecture in elite soccer players. *Journal of Sports Medicine and Physical Fitness*, 56(11), 1269–1278.
- Spinetti, J., Figueiredo, T., Willardson, J., Bastos de Oliveira, V., Assis, M., Fernandes de Oliveira, L., Miranda, H., Machado de Ribeiro Reis, V. M., & Simão, R. (2019). Comparison between traditional strength training and complex contrast training on soccer players. *The Journal of Sports Medicine and Physical Fitness*, 59(1), 42–49. <https://doi.org/10.23736/S0022-4707.18.07934-3>
- Suarez-Arrones, L., Saez De Villarreal, E., Nunez, F. J., Di Salvo, V., Petri, C., Buccolini, A., Maldonado, R. A., Torreno, N., & Mendez-Villanueva, A. (2018). In-season eccentric-overload training in elite soccer players: Effects on body composition, strength and sprint performance. *PloS One*, 13(10), e0205332. <https://doi.org/10.1371/journal.pone.0205332>
- Suchomel, T. J., Comfort, P., & Lake, J. P. (2017). Enhancing the force-velocity profile of athletes using weightlifting derivatives. *Strength & Conditioning Journal*, 39(1), 10. <https://doi.org/10.1519/SSC.0000000000000275>
- Suchomel, T. J., Comfort, P., & Stone, M. H. (2015). Weightlifting pulling derivatives: Rationale for implementation and application. *Sports Medicine (Auckland, N.Z.)*, 45(6), 823–839. <https://doi.org/10.1007/s40279-015-0314-y>
- Suchomel, T. J., McKeever, S. M., McMahon, J. J., & Comfort, P. (2020). The effect of training with weightlifting catching or pulling derivatives on squat jump and countermovement jump force–time adaptations. *Journal of Functional Morphology and Kinesiology*, 5(2), 28. <https://doi.org/10.3390/jfmk5020028>

- Suchomel, T. J., Nimphius, S., & Stone, M. H. (2016). The importance of muscular strength in athletic performance. *Sports Medicine (Auckland, N.Z.)*, 46(10), 1419–1449. <https://doi.org/10.1007/s40279-016-0486-0>
- Talpey, S. W., Young, W. B., & Saunders, N. (2014). The acute effects of conventional, complex, and contrast protocols on lower-body power. *Journal of Strength and Conditioning Research*, 28(2), 361–366. <https://doi.org/10.1519/JSC.0b013e318299a68b>
- Talpey, S. W., Young, W. B., & Saunders, N. (2016). Is nine weeks of complex training effective for improving lower body strength, explosive muscle function, sprint and jumping performance? *International Journal of Sports Science & Coaching*, 11(5), 736–745. <https://doi.org/10.1177/1747954116667112>
- Thapa, R. K., Lum, D., Moran, J., & Ramirez-Campillo, R. (2021). Effects of complex training on sprint, jump, and change of direction ability of soccer players: A systematic review and meta-Analysis. *Frontiers in Psychology*, 11, Article 627869. <https://doi.org/10.3389/fpsyg.2020.627869>
- Turner, A. N., Comfort, P., McMahon, J., Bishop, C., Chavda, S., Read, P., Mundy, P., & Lake, J. (2020). Developing powerful athletes, Part 1: Mechanical underpinnings. *Strength & Conditioning Journal*, 42(3), 30–39. <https://doi.org/10.1519/SSC.0000000000000543>
- Wagle, J. P., Taber, C. B., Cunanan, A. J., Bingham, G. E., Carroll, K. M., DeWeese, B. H., Sato, K., & Stone, M. H. (2017). Accentuated eccentric loading for training and performance: A review. *Sports Medicine (Auckland, N.Z.)*, 47(12), 2473–2495. <https://doi.org/10.1007/s40279-017-0755-6>
- Weakley, J., Mann, B., Banyard, H., McLaren, S., Scott, T., & Garcia-Ramos, A. (2021). Velocity-based training: From theory to application. *Strength & Conditioning Journal*, 43(2), 31–49. <https://doi.org/10.1519/SSC.0000000000000560>
- Weakley, J., Wilson, K. M., Till, K., Read, D. B., Darrall-Jones, J., Roe, G. A. B., Phibbs, P. J., & Jones, B. (2019). Visual feedback attenuates mean concentric barbell velocity loss and improves motivation, competitiveness, and perceived workload in male adolescent athletes. *Journal of Strength and Conditioning Research*, 33(9), 2420–2425. <https://doi.org/10.1519/JSC.0000000000002133>
- Wilson, J. M., Marin, P. J., Rhea, M. R., Wilson, S. M. C., Loenneke, J. P., & Anderson, J. C. (2012). Concurrent training: A meta-analysis examining interference of aerobic and resistance exercises. *Journal of Strength and Conditioning Research*, 26(8), 2293–2307. <https://doi.org/10.1519/JSC.0b013e31823a3e2d>
- Winter, E. M., Abt, G., Brookes, F. B. C., Challis, J. H., Fowler, N. E., Knudson, D. V., Knuttgen, H. G., Kraemer, W. J., Lane, A. M., van Mechelen, W., Morton, R. H., Newton, R. U., Williams, C., & Yeadon, M. R. (2016). Misuse of “power” and other mechanical terms in sport and exercise science research. *Journal of Strength and Conditioning Research*, 30(1), 292–300. <https://doi.org/10.1519/JSC.0000000000001101>
- Wisløff, U., Castagna, C., Helgerud, J., Jones, R., & Hoff, J. (2004). Strong correlation of maximal squat strength with sprint performance and vertical jump height in elite soccer players. *British Journal of Sports Medicine*, 38(3), 285–288. <https://doi.org/10.1136/bjsm.2002.002071>
- Zimmermann, H. B., Knihs, D., Diefenthaler, F., MacIntosh, B., & Pupo, J. D. (2021). Continuous jumps enhance twitch peak torque and sprint performance in highly trained sprint athletes. *International Journal of Sports Physiology and Performance*, 16(4), 565–572. <https://doi.org/10.1123/ijsp.2020-0240>